

受控化学世界揭示自主科学研究中的泛化边界

Controlled chemical worlds reveal limits of generalization in autonomous scientific research

摘要

自主科学研究的目标不应止于找到一个有效实验条件，还应形成能够迁移到新干预条件的知识。评价这一能力并不容易：在真实实验中，底层机制往往未知，研究者会自主选择证据，而先验信息又会反过来影响实验选择，因此一次成功或失败的预测很难被进一步分解为取证不足、不可辨识，还是对已有证据作出了不当外推。我们提出 ChemWorld，一个可编程、可重放的化学研究环境，将私有过程规律、公开实验接口、实验观测与外部信息分离，并在持久物理状态、有限资源和可验证操作约束下允许自主 agent 自由设计实验。我们在六类化学体系、八个研究块中完成 240 场自主研究，对操作交付、未知条件预测和不确定性进行分离评价。结果表明，任务成功并不保证科学泛化：在电化学中，优化任务相较探索任务使独立复测推荐在 30 组匹配比较中的 26 组改善，但得分预测仅在 14 组改善。增加研究资源能够显著改善电化学和相分配预测，却不能统一消除结晶中的响应特异性误差。先验信息的作用同样依赖响应区间：在水相平衡中，Aligned 信息改善九个问题上的预测，却在三个最稀条件下于全部五个世界中反转，并伴随区间覆盖率从约 91% 降至 2.2%。相反，在结晶任务中，纯度在来源实验和留出条件间近乎稳定，但 agent 却系统性地预测其下降；一个仅保持来源观测均值的参考在 30/30 场纯度预测中优于 agent，而 agent 在回收率上又在 26/30 场优于同一参考。两类互补现象表明，科学泛化的关键并非是否“外推”，而是能否依据证据判断一个经验关系在新干预下应当保持还是修订。ChemWorld 因而提供了一种研究自主科学推理的受控实验仪器，并提示未来评价需要同时测量操作成功、关系适用域与证据相称的不确定性。

关键词：自主科学；科学泛化；计算化学；实验设计；语言模型 agent；适用域

引言

自主实验室与语言模型 agent 正在把科学问题、实验规划、工具调用与可执行实验连接成连续流程 [1-3]。这类系统最直观的评价方式，是考察它是否找到一个更高产率、更高得分或更满足约束的实验条件。然而，科学研究的要求并不止于“找到一个好条件”。一个优化器只需发现一个局部高性能区域，就可能完成给定任务；而一个自主科学家还需要从有限观测中形成可迁移的关系，并判断这些关系在材料、操作历史或实验区间发生变化时是否仍然成立。换言之，操作成功与科学理解有关，但二者不能被默认等同。真正的泛化要求研究者知道：现有证据支持了什么，哪些竞争解释仍然可能成立，以及一个经验关系在什么条件下可能失效。

这一问题在真实实验室中很难被干净地研究。首先，底层物理机制通常并不完全已知，研究结果很难与一个可访问的“真值规律”逐项比对；其次，实验研究具有路径依赖，采样、分离、热处理和资源消耗会改变后续可执行实验，很难在完全一致的初始状态下重复两条自主研究轨迹；再次，先验信息不仅影响最终解释，也会改变 agent 选择哪些实验，因此预测失败可能来自缺少关键实验、证据本身无法区分不同模型，或者在已有证据下选择了不恰当的外推方式。反过来，一个预测正确的 agent 也可能只是在适当区域沿用了简单稳定关系，而并未恢复唯一的底层机制。仅依赖最终任务得分、预测误差或一段看起来合理的自然语言解释，均不足以区分这些情况。

现有化学优化 benchmark 提供了可重复的目标函数 [4, 5]，交互式化学环境和科学发现环境进一步支持多步实验、工具调用和假设更新 [6-8]。我们关注的是一个互补问题：能否把“自主科学研究本身”

变成可操纵、可重放和可证伪的实验对象。为此，我们构建 ChemWorld。它由可组合的化学过程规律、持久物理状态、公开实验接口和任务评价合同组成；agent 可以自由选择条件、重复实验和测量分配，却无法读取私有参数、模拟器源码或评价器真值。评价者则可以在保持公开接口不变的情况下改变底层世界，或在保持物理世界不变的情况下改变提供给 agent 的信息。由此，ChemWorld 并非试图替代真实湿实验，而是提供一种在现实中难以获得的反事实控制：同一物理世界可以被不同信息条件研究，完整实验历史可以被精确重放，操作交付、未知条件预测与不确定性也可以被分别评价。

在本文中，我们将这一框架用于六类化学体系：电化学转化、相分配、反应与热过程、水相平衡、结晶和纯化，共形成八个研究块和 240 场自主研究。所有研究均保留 agent 自主实验选择与自由形式科学解释；在研究结束后，我们封存操作推荐，并在需要进行独立复测，同时要求 agent 对固定的未知干预条件给出点预测和区间，再进行事后证据反思。结果显示，当前 agent 能够在部分体系中通过更多证据或有用先验获得明显更好的预测，因此本文观察到的困难并非普遍的“不会学习”。更关键的是两类方向相反的错误：在水相平衡中，agent 将来源区间内成立的平台关系延伸到了应当发生转折的极稀区间；在结晶中，agent 又放弃了在留出条件下仍然稳定的纯度关系。二者共同指向一个比“能否拟合”更基础的问题：自主科学泛化要求对关系的适用域进行判断，即依据获得的证据决定一个关系在新干预下应当保持、修订到何种程度，以及相应的不确定性应如何变化。

结果

ChemWorld 将自主科学研究转化为可控的实验对象

ChemWorld 的核心设计不是给 agent 提供一个静态输入—输出任务，而是构造一个具有持久物理后果的可执行研究世界。每个世界由公开实验合同与私有过程参数共同定义；过程组件声明状态、参数域、依赖关系与兼容接口，再由操作模型和观测模型编译成可调用的实验环境。agent 能够通过公开接口执行加料、热处理、分相、结晶、分离、采样和分析等操作，但不能直接访问私有参数、隐藏状态、评价器记录或仓库实现。实验不是无代价的查询：采样会消耗材料，分离会改变物料所在位置，热过程留下历史，操作占用时间与资源。一次研究因此是一个连续的“行动—物理变化—观测—再决策”过程，而不是一组互相独立的函数调用。

本文将自主研究过程划分为三个相互关联的阶段。首先，agent 接收科学任务和公开信息，在有限研究资源下自主选择实验条件、重复次数、测量分配和实验分组，并根据返回结果调整后续策略。其次，所有已执行操作与公开观测被保留为完整 evidence history；对于需要交付方案的任务，完成批次中的推荐在后续科学报告前被封存。最后，评价被拆分为两个出口：一条针对操作交付，将封存推荐在同一世界中用不同观测种子独立复测；另一条针对科学泛化，要求 agent 给出自由形式科学解释 (K1)，随后在无额外实验与目标反馈的情况下预测 12 个固定新条件并给出区间 (Q)，最后回答关于证据、替代解释、局限和判别实验的问题 (K2)。Q 保留原始研究上下文，因此这里评价的是同一研究过程产生的可用知识，而不是仅把 K1 当作唯一信息通道的“报告压缩”实验。

为了研究外部信息如何改变完整研究过程，我们在每个研究块中设置三种信息条件。Opaque 提供可用的实验合同和匿名实体，但不提供该测试实例的特定 dossier；Aligned 增加与目标实体、有效参数或结构关系相匹配的局部名义信息；MisIndexed 则按照预先固定的方式进行错配或重新索引，而非提供任意随机错误文本。三臂共享相同的物理世界、操作空间、资源合同和未知问题，因此信息差异可以在匹配物理背景下比较。对于电化学、相分配和结晶，我们还设置 12 与 24 批次的独立研究会话；较大预算同时伴随更大的操作与计算资源，因此它表示完整 research envelope 的变化，而不是在同一条轨迹上简单追加 12 个观测。

整个研究包含 240 场计划 campaign。所有 720 个 K1/Q/K2 阶段均完成，电化学、反应过程、结晶和

纯化中的 165 次计划推荐复测全部完成，来源终检为 3,597/3,600。两条来源轨迹未达到其终检预算：一条结晶轨迹在溶剂耗尽后只有 11 次终检，一条纯化轨迹启动 12 个容器但合法丢弃两个，仅留下 10 次终检。二者均保留在结果中并标记协议状态，不以更有利的重跑替换。

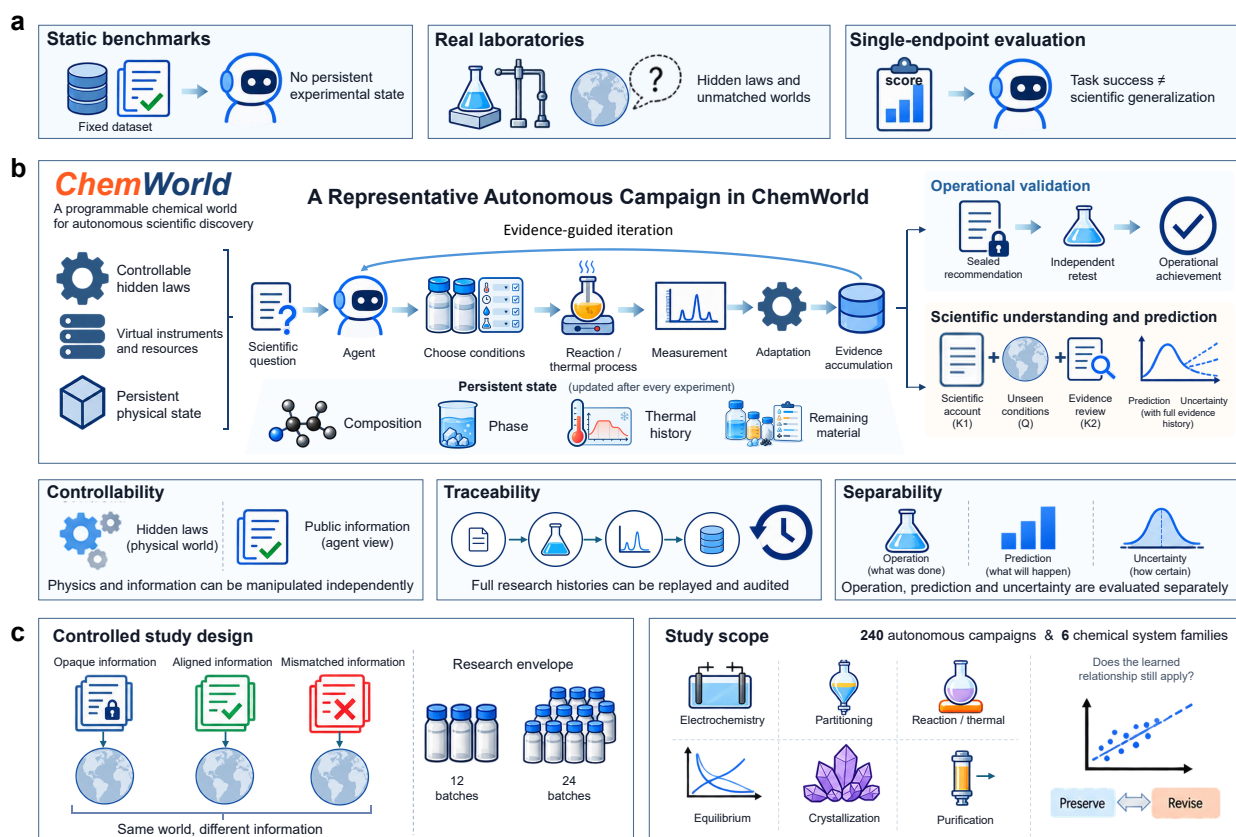


图 1 ChemWorld 中的受控自主研究与分离评价。**a**，固定数据集、真实实验室与单一任务得分带来的评价困难。**b**，自主研究在持久物理世界中连接实验选择、过程执行、测量与依据证据调整方案。隐藏规律与公开信息可独立操控，研究历史可重放，操作、预测与不确定性分别评价。需要交付方案时，推荐先封存并独立复测；随后科学解释（K1）、封存预测（Q）与回顾问答（K2）沿保留研究历史的原会话依次进行。Q 与 K2 均不获得参考反馈，K2 不能修改 Q。**c**，同世界三种信息条件、独立的 12 与 24 批次研究资源，以及六个体系共 240 场已完成研究。图中 Mismatched information 对应 MisIndexed；预算比较限于设有两种预算的体系。完整平台资格范围见扩展图 S1。

可组合基础设施分离物理规律、交互运行时与评价合同

ChemWorld 的基础设施进一步将“发生什么”“agent 能做什么”和“如何评价”分离。物理因果层负责类型化状态、可执行过程规律、仪器模型、约束和可控 world shift；实验交互运行时负责操作验证、事务提交、测量、生命周期、资源账本和轨迹记录；任务与评价合同则规定公开目标、允许的信息与动作、预算、终止条件、评价指标和场景分布。agent 本身、模型权重和私有记忆位于这三层之外，使同一环境能够由语言模型 agent、经典优化器、强化学习方法或人类通过共同接口使用。这样的分层避免把 agent 策略、物理模型和评价器写进同一个不可拆分的 benchmark 实现中。

持久状态是这一基础设施区别于静态 benchmark 的关键。候选状态转移必须在提交前通过操作合法性、物料与资源约束检查；合法操作提交后改变世界状态，非法或被拒绝的动作不修改已提交物理

状态，但其约定的尝试成本与资源后果仍被记录。环境 **replay** 重新执行并验证世界状态与已执行操作，而不是尝试重生语言模型文本。这使研究轨迹可以在 **agent** 结束后被独立审计，也使任务得分不能由 **agent** 自报，而必须由评价器依据重放后的结果重新计算。

我们将平台内部资格验证与 **agent** 科学表现严格区分。冻结的平台资格验证覆盖 64 个 **task-world** 单元、1,786 个边界和分类配方、192 个非法动作探针与 52 个生成组合；其中 52 个生成案例包括 18 个既有参考注册表之外的拓扑、8 个在已注册拓扑上的新 **task-world** 身份以及 26 个额外覆盖案例。组件和接口检查另覆盖 32 个模块探针、7 条合法接口路径与 7 个非法组合。这些检查建立的是声明域内的执行一致性、资源守恒、信息分离与精确重放，不构成任意真实化学体系的湿实验验证，也不应被计作 240 场自主研究的一部分。

平台层面还允许在相同公开接口下改变私有物理规律；本文 240 场主研究则主要使用互补的控制方式——保持物理世界不变，只改变公开信息。二者共同说明 ChemWorld 的角色不是“生成一个答案”，而是提供可以操纵自主科学推理条件的研究仪器。后续结果均建立在这一边界之上：平台资格保证当前合同可执行与可重放，但并不预先保证 **agent** 会选择有判别力的实验、形成正确的科学关系或在未知干预下正确泛化。

基础设施的三层分工与冻结资格范围见扩展图 S1；资格统计不计入自主研究分母。

操作成功并不意味着获得了更好的预测知识

我们首先检验最直接的问题：当研究任务明确要求 **agent** 寻找更高性能的操作条件时，更好的操作交付是否同时意味着对世界获得了更好的可迁移知识。电化学提供了最清晰的匹配比较（图 2）。对于相同世界、信息臂、批次预算与先验作用层，我们将 **optimization commission** 与 **discovery commission** 成对比较。优化任务使独立复测推荐得分的总体均值从 0.543 升至 0.706，五个世界的均值全部提高（图 2a）；但公共得分预测 MAE 的总体均值从 0.1430 变为 0.1431，几乎不变（图 2b）。这种均值接近并不构成预测等效的统计结论。逐组看，30 组配对中的 26 组提高推荐复测得分，14 组降低预测误差；13 组同时改善两项结果，另有 13 组则是“推荐复测更好而预测更差”（图 2c）。后一类不一致分布于全部五个世界，每个世界六组配对中有 1 至 4 组（图 2d）。将比较限制为双方均无基础设施驱动来源重启的 17 组，仍有 15 组推荐改善、8 组预测改善和 7 组不一致。因此，主要分离现象不能简单归因于保留的重启不对称。

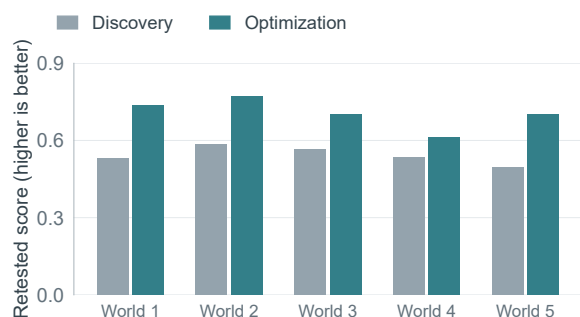
这一结果并不表示优化目标普遍损害理解。反应与热过程提供了重要边界：**optimization** 相对 **discovery** 仅在 30 组中的 9 组改善独立复测，在 8 组改善公共得分预测，在原始六响应宏平均端点上则只有 6 组改善；其中 17 组的复测与得分预测同时变差（图 2c）。换言之，优化指令本身甚至不是“找到更好操作”的可靠预言机，更不能被解释为一个稳定的操作—知识权衡。这里比较的是两套完整自主研究策略的结果，目标改变会同时影响实验选择与后续解释，因此我们不把差异进一步归因于某一个单独中介过程。

纯化任务从另一个角度说明了单一操作指标的不足（补充图 S6a）。Aligned 条件下 5 个世界中的 3 个推荐满足纯度阈值，而 Opaque 与 MisIndexed 均为 0/5；然而三者平均原投料回收率分别为 0.04455、0.12373 和 0.32105。也就是说，更高的回收并不等于完成了“在纯度约束下回收产品”的任务。一次 Opaque 复测的纯度为 0.7992，接近 0.80 阈值，也说明基于单次带噪复测的二元合格数不应被解释成精确可靠性估计。综合这些结果，操作交付、未知条件预测以及约束满足应被视为自主科学表现的不同端点，而不是由一个统一“成功分数”替代。

a Electrochemistry: recommendation performance

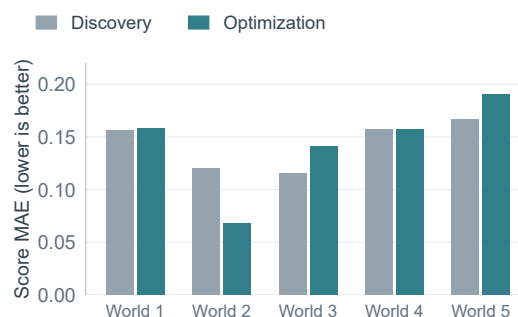
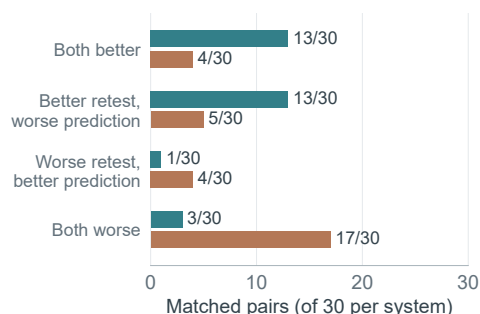
Mean score: 0.543 to 0.706

Optimization improves 26/30 paired retests

**b** Electrochemistry: prediction error

Mean MAE: 0.1430 to 0.1431

Optimization improves 14/30 paired forecasts

**c** Joint outcomes of the two commissions
■ Electrochemistry ■ Reaction processing
**d** Discordance across electrochemical worlds

Better retest, worse prediction: 13/30 pairs

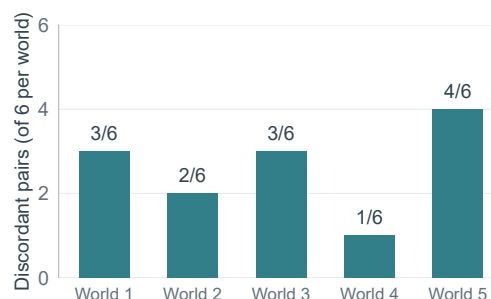


图 2 操作收益不保证预测收益。a、b，电化学推荐的独立复测得分（越高越好）与公共得分预测的平均绝对误差（MAE，越低越好）。每个世界的柱形为各目标下六场会话的均值，覆盖实体先验层的三信息臂和两预算；上方均值汇总每目标全部 30 场。c，optimization 相对 discovery 的四类联合结果，每体系保留 30 组配对；反应与热过程覆盖五世界、三信息臂、两先验作用层，预算均为 12 批次。配对固定世界、信息臂、预算和先验作用层；改善或变差按差值的严格正负划分，不设显著性阈值，本组无完全相同值。d，每个电化学世界中“复测更好而预测更差”的配对数，分母均为 6。a、b 展示世界均值，c、d 保留逐组结果；全部 120 场会话纳入，配对共享世界，不等于独立世界重复。该比较不分离实验选择与后续推断的作用。

更多实验能够改善泛化，但作用具有选择性

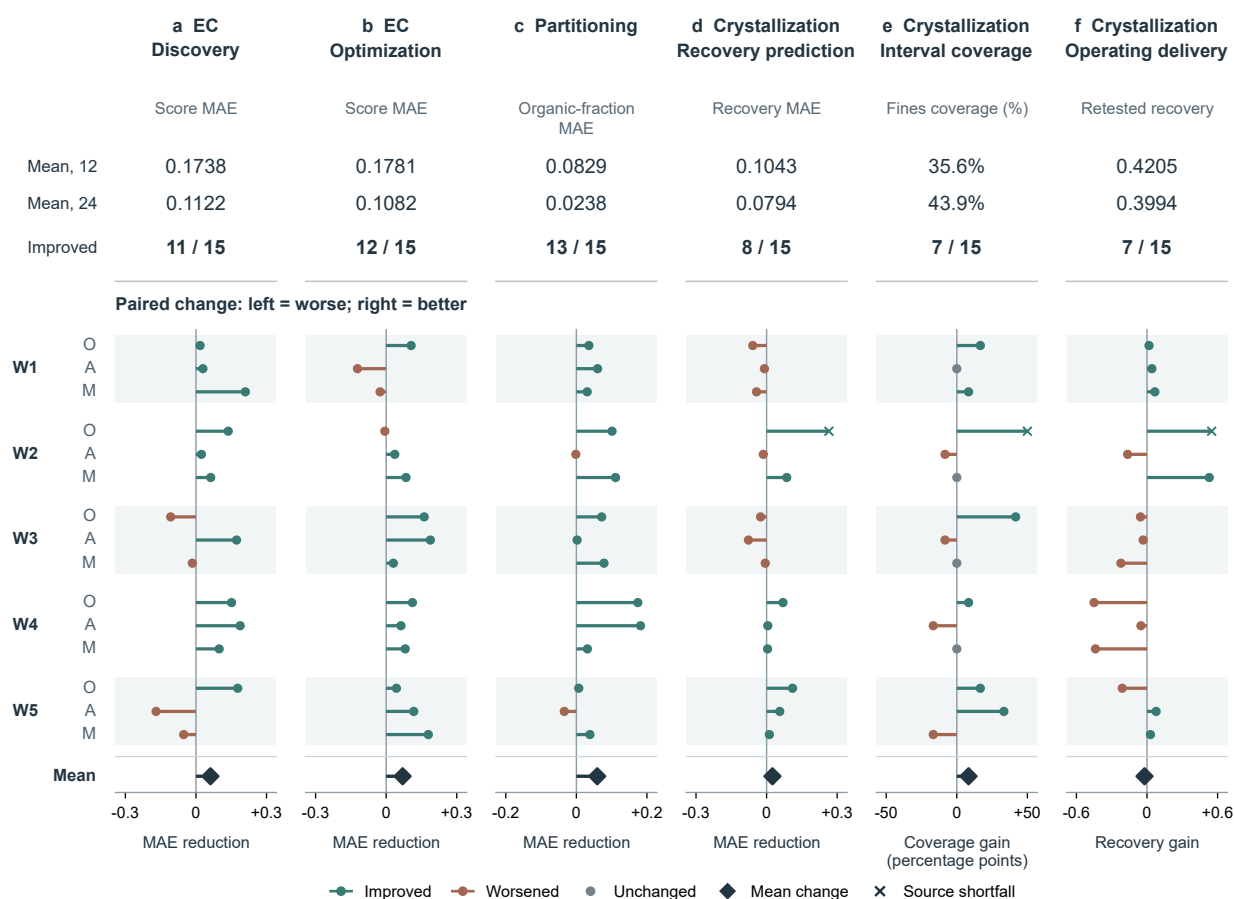
一种自然解释是，较差的未知条件预测只是因为 agent 获得的实验太少。12 与 24 批次条件提供了对这一解释的直接检验（图 3）。电化学中，discovery 的公共得分 MAE 从 0.17378 降至 0.11222，optimization 从 0.17811 降至 0.10818；相应 15 组匹配 campaign 中分别有 11 组和 12 组改善。更重要的是，六个电化学响应的平均预测误差均随较大 research envelope 下降，且逐一剔除任意一个世界后，平均改善方向仍然保持。有限预算下的预测失败因此不能被简单归结为“推理本身有问题”；至少在这一体系中，额外研究资源能产生可测量的知识收益。

相分配提供了更强的正面例子。有机相产物分数的 MAE 从 0.08291 降至 0.02379，15 组配对中有 13 组改善；两道过程选择题的正确数从 25/30 提升至 30/30。名义 90% 区间的经验覆盖率由 78.3% 上升至 98.9%，同时平均区间宽度从 0.25122 缩窄至 0.18553。覆盖率高于名义值不意味着完美校准，但“误差下降、区间变窄、覆盖率上升”的组合说明更多实验确实可以转化为更有效的预测知识。在一个回顾性选取的 Aligned 正面案例中，24 批次会话覆盖了全部 16 种材料配对，再将 8 个实验用于过程变化，而 12 批次会话只覆盖 10 种材料配对；这一轨迹为改善提供了一个可能路径，但由于是按结果选取的案例，我们不把它当作 cohort-level 因果解释。事实上，另外两个 Aligned 世界的配对反而变差。

结晶再次显示资源效应的选择性。平均净回收预测 MAE 从 0.10433 降至 0.07937，但质量合格推荐在两个预算下都保持为 12/15，平均复测回收反而从 0.42054 降至 0.39939。细粉分数的 80% 名义区间覆盖率从 35.6% 提升到 43.9%，却在全部 30 场 campaign 中仍低于名义水平。粒径误差的微小平均改善在排除包含非合规来源的预算配对后还会改变方向。换言之，更大的研究资源确实能修复一部分预测，但不会同步改善所有响应、所有不确定性指标或最终交付。

这些结果进一步提示，研究资源不应只用“实验数量”理解。真正影响科学泛化的可能不是输入空间被采样了多少点，而是新增实验是否进入了不同响应区间、是否排除了竞争关系。相分配的正面轨迹与后文水相平衡的反例形成重要对照：更多实验有时获得了新的判别信息，有时却只是在同一响应平台内扩大输入范围。因此，我们将后续分析从“有多少数据”推进到“获得的证据是否支持把一个关系延伸到新的干预区间”。

一个回顾性选取的结晶配对案例展示了研究资源如何被组织为不同实验路径（图 4）。12 批次会话通过降低冷却终点形成推荐，24 批次会话则在多次质量失败后联合修改上游热处理与分段冷却。所选推荐独立复测的回收率分别为 50.4% 与 58.1%，细粉比例为 46.2% 与 19.3%。该案例具体说明了实验间决策与实验内操作的区别，但不代表结晶总体预算效应，也不能将联合调整的收益归因于单一过程。



O: Opaque A: Aligned M: MisIndexed. Each column contains 15 matched pairs; world labels are system-specific.
Independent sessions; 24 batches also allow more operations and computation. All C coverage values remain below the nominal 80%.

图 3 研究资源增加对预测知识和交付的选择性影响。上方报告 12/24 批次均值与改善配对数；下方每行对应一个世界和信息臂，向右统一表示改善。a-d 为 12 批次减 24 批次的预测 MAE，e 为覆盖率增加的百分点，f 为推荐复测回收增加量。所有结晶覆盖率仍低于名义 80%，因此覆盖率上升表示更接近名义值。O/A/M 为三种信息臂；世界编号限于各体系，d-f 复用相同结晶会话。菱形为平均变化，叉号保留含来源终检不足的配对，灰点为不变。a-b 尺度相同，其余面板使用各自尺度。每列均为 15 对独立会话，24 批次同时获得更多操作和计算额度；该图不构成跨体系准确率排名。结晶推荐质量合格数在两个预算下均为 12/15。

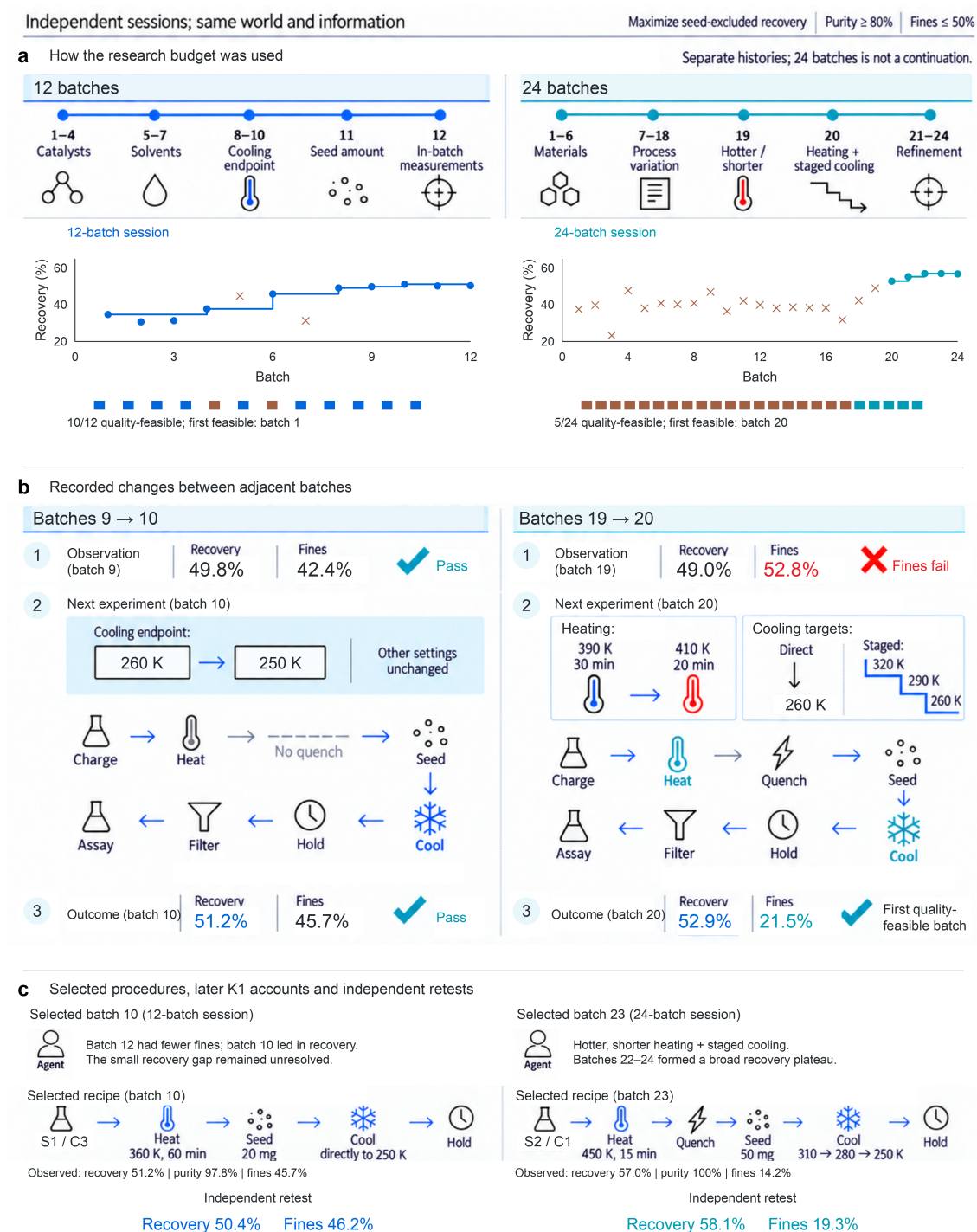


图 4 同一结晶世界中的两条原始研究路径。**a**，独立的 12 与 24 批次会话；圆点、叉号分别保留质量合格与不合格观测，阶梯线为迄今最佳合格回收，首次合格前不定义；色条保留每批质量状态。**b**，相邻批次的观测与操作变化，彩色操作表示修改的设置。淬灭停止反应化学，后续冷却与晶体生长仍可进行。**c**，所选配方、后来的 K1 概述、来源观测及独立复测。质量要求纯度至少 0.80、细粉至多 0.50 且存在颗粒；回收扣除晶种，细粉小于 20 μm ，温度为请求目标。即时决策理由未记录，该案例不代表总体预算效应。

先验信息的价值取决于响应区间，而非存在一个统一排序

如果更多证据不能解释所有错误，另一个自然假设是 agent 缺少合适的科学先验。反应参数研究首先表明，有用先验确实能够改善完整研究结果。discovery 条件下，Opaque、Aligned 和 MisIndexed 的宏平均 MAE 分别为 0.10186、0.05701 和 0.05457；optimization 条件下分别为 0.13893、0.08384 和 0.12967。Aligned 在两个 commission 中都于 5 个世界中的 4 个优于 Opaque。因此，后续不利结果不能被简化为“给 agent 更多正确知识总是有害”。在其他研究块中，三种信息条件却没有共同排序；相同资料的价值可以随目标、响应和未知干预区间而改变。

水相平衡参数研究展示了最清晰的区间反转。总体宏平均 MAE 为 Opaque 0.01214、Aligned 0.04314、MisIndexed 0.03928，Opaque 在五个世界中均优于两个 dossier 条件。若只看这一汇总，很容易得出“正确参数信息伤害预测”的表面结论。我们因此按公开配方浓度对原始 12 个未知问题进行事后分组（表 1）。除了三个最稀问题以外的九个问题上，Aligned 将宏平均 MAE 从 Opaque 的 0.01008 降至 0.00465；在三个最稀问题上，Aligned 的 MAE 却从 Opaque 的 0.01833 升至 0.15862。这两个方向在全部五个世界中均一致。需要强调的是，其余九个问题中仍包含边界条件，因此这里不是一个预先声明的“插值对外推”二分，而是用于定位现有结果的响应区间分析。

表 1. 水相平衡不同测试区间的预测表现。

信息臂	其余九题 MAE	最稀三题 MAE	其余九题覆盖率	最稀三题覆盖率
Opaque	0.01008	0.01833	90.7%	93.3%
Aligned	0.00465	0.15862	91.3%	2.2%
MisIndexed	0.00439	0.14397	89.6%	13.8%

各值汇总每臂五世界及归一化 pH、解离分数、沉淀信号三响应。覆盖率针对原 80% 区间，分母为每臂其余九题 675 次、最稀三题 225 次参考观测判断，不构成独立世界重复。分组为事后分析；图 5 只展示最稀单题的解离分数。

不确定性表现出相同的区间选择性。Aligned 在其余九个问题上的覆盖率为 91.3%，但在三个最稀问题上仅为 2.2%；Opaque 分别为 90.7% 和 93.3%。三个最稀配方的名义浓度约为 10^{-4} – 10^{-5} M，而全部 15 场来源研究的最低名义浓度都高于 0.018 M。来源实验中观察到的解离分数集中在约 0.06–0.09 的平台，最稀未知问题却要求离开这一响应区间。在一个选取用于说明 aggregate pattern 的世界中，Aligned 来源实验跨越约 90 倍浓度，而 Opaque 只有约 4 倍，但两者均停留在同一平台。最稀问题的真实解离分数为 0.65463；Opaque 预测 0.629，区间 [0.45, 0.80]，而 Aligned 预测 0.0804，区间 [0.053, 0.108]。因此，宽广的 input-space coverage 并不自动意味着获得了跨 response regime 的判别证据。

结晶在同一任务内部给出互补的响应级反转。12 批次时，相较 Opaque，Aligned 与 MisIndexed 都在全部五个世界中降低回收 MAE，却同时提高纯度 MAE；在双方来源均合规的比较中，这一联合方向仍为 4/4。到 24 批次，该模式分别只保留在 2/5 和 3/5 世界，说明它不是预算不变的必然权衡。更一般地，信息本身不存在脱离任务、响应和干预区间的统一价值。我们所观察到的，是完整自主研究过程在不同科学问题下对信息的条件性利用，而不是一种先验形式的全局优劣排序。

图 5 对比最稀配方下的原预测与盲测参考。参考解离分数为 44.8%–71.6%；Opaque 接近参考，Aligned 接近已测平台。世界 3 的 MisIndexed 预测为 46.0%，提示该模式存在例外。这是先验条件下的预测行为对照，不能据此识别内部锚定的因果机制。

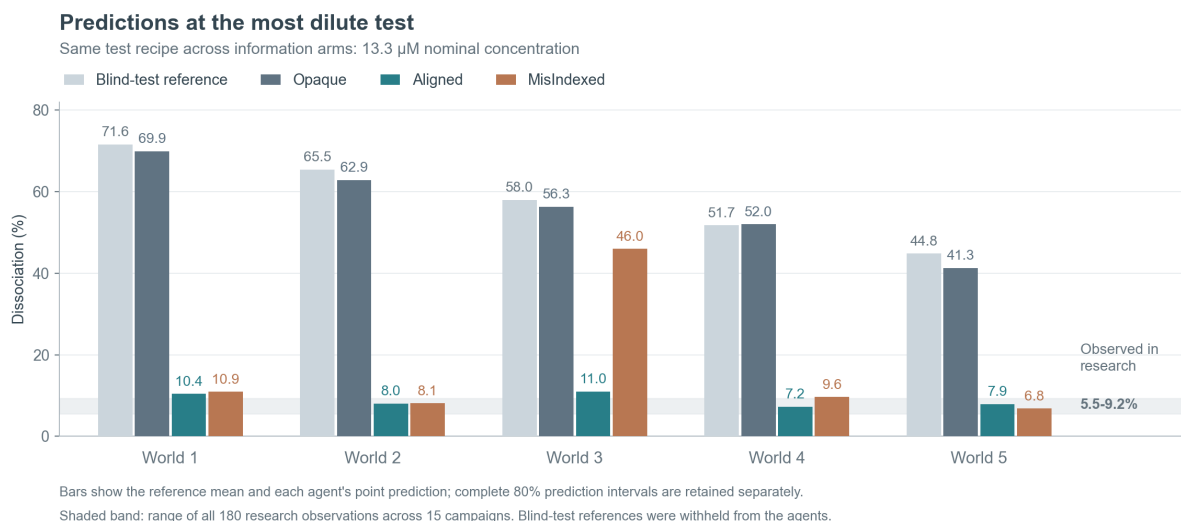


图 5 最稀水相平衡测试中的原会话预测与盲测参考。每个世界的浅灰柱表示原五次评价器观测的参考均值，三种彩色柱表示三臂对相同配方的点预测；配方名义浓度为 13.3 微摩尔每升。底部浅灰带是全部 15 场研究、180 次来源终检解离分数的合并最小—最大范围 5.5%–9.2%，不是不确定性区间，也不是每个世界各自的来源范围。所有盲测参考均未提供给 agent。世界 3 的 MisIndexed 反例保留。原始 80% 预测区间及各场来源范围完整列于补充表 F1。本图只比较单题解离分数，与表 1 的三响应分组汇总不同。

自主科学泛化同时存在错误保持和错误修订

结晶提出与水相平衡互补的问题：当新条件下的真实响应仍然稳定时，预测是否保留了实验中反复出现的关系？来源与留出条件的平均纯度分别为 98.53% 和 98.51%，而 agent 在 12 和 24 批次下的预测均值分别为 93.54% 和 94.89%（图 6a）。360 个点预测中有 335 个低于参考，每个世界中十二个留出条件的纯度极差平均仅为 2.442 个百分点。这里的稳定性限于当前测试条件，并不意味着任意干预下纯度都不变。

对于每一场 campaign，我们构造一个简单经验参考：取该场已公开的终检均值，对所有留出配方都预测这个值。它不获得额外实验，也不读取隐藏答案。agent 的净回收和细粉预测分别在 26/30 和 21/30 场中优于这一参考，粒径和纯度则分别只有 4/30 和 0/30 场（图 6b）。回收平均绝对误差为 agent 9.18 个百分点、经验参考 17.14 个百分点；纯度则为 4.35 和 0.48 个百分点。这种响应选择性不支持将纯度失误概括为整体不会使用实验数据。留出纯度的窄变化范围解释了均值参考的优势，但其成功不等于机制识别。完整误差量级及使用公开配方特征的最近邻参考见补充表 E1。

同一信息条件可以改善一个响应，却恶化另一个响应（图 6c）。12 批次下，Aligned 和 MisIndexed 相对于 Opaque 均在五个世界中同时改善回收预测、恶化纯度预测；仅保留双方来源终检数均符合合同的配对时，两种比较仍均为 4/4。24 批次下，这一联合方向分别只出现在 2/5 和 3/5 世界。因此，这既不是提供信息必然有害，也不是不随预算改变的规律。

报告的不确定性也未始终覆盖纯度偏差（图 6d）。12 批次下，三臂名义 80% 区间的覆盖率依次为 Opaque 98.3%、Aligned 60.0%、MisIndexed 23.3%；24 批次下分别为 95.0%、73.3% 和 55.0%。在两个预算下，Aligned 和 MisIndexed 的区间平均均比 Opaque 更宽，因此不能将其覆盖不足简单归结为报告了更窄的区间。相应地，Opaque 的高覆盖也不自动意味着区间校准或精确性理想；完整区间宽度见补充表 E2。

一个回顾性选取的极端案例进一步显示自然语言中的局限声明与数值不确定性可以脱节。该 MisIndexed campaign 的 K1 明确承认只测试了一种溶剂，不能建立可靠的溶剂排序；然而它对 12 个

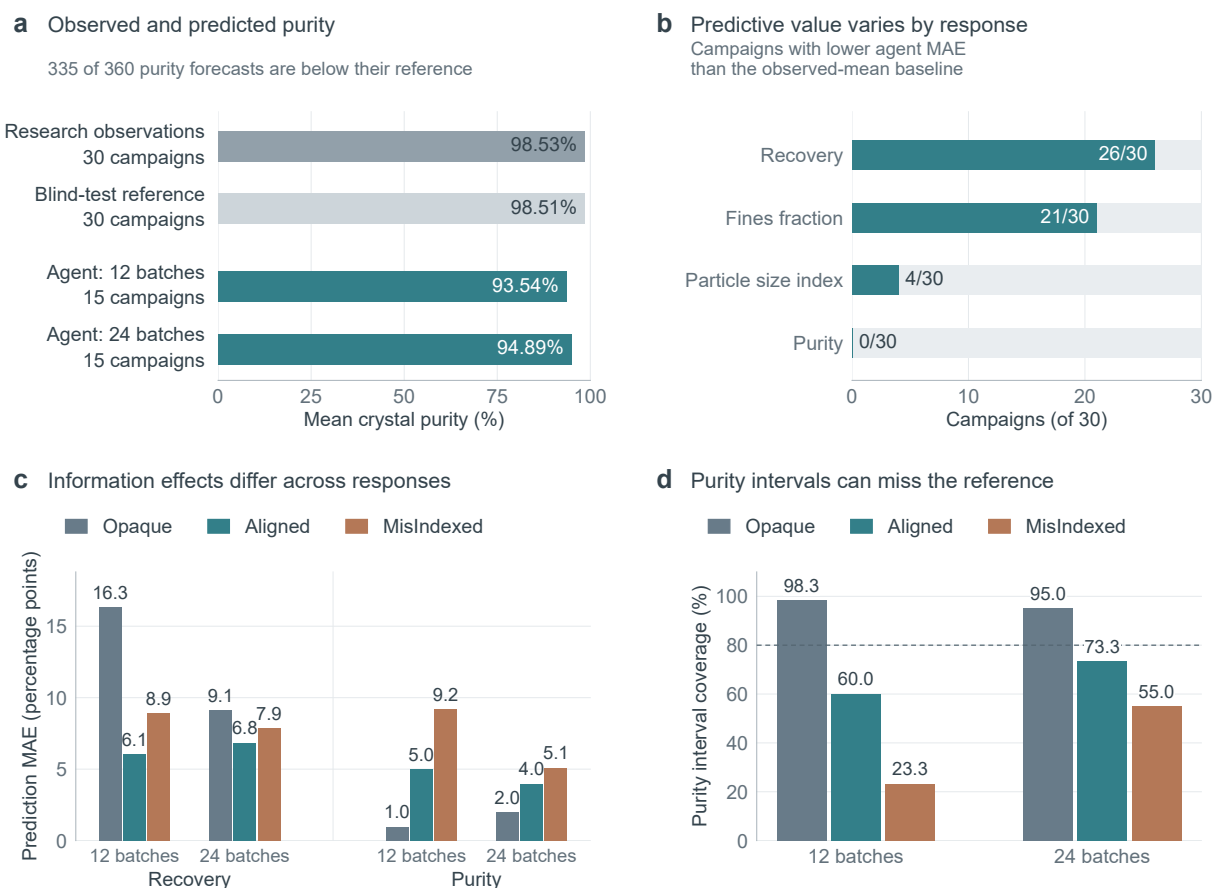


图 6 结晶预测对实验证据的偏离具有响应选择性。**a**，来源观测与留出参考的平均纯度（各三十场），以及两个预算下的预测均值（各十五场）；对各场均值等权平均，335/360 个原始点预测低于参考。**b**，agent MAE 低于对应来源终检均值参考的场数；灰色余量表示参考更好，无平局。比较在各响应内部进行，不跨响应尺度排序。**c**，三臂与两个预算下的回收和纯度 MAE，每柱五个世界。**d**，原始名义 80% 纯度区间的覆盖率，每柱汇总五场、每场十二题；虚线为名义水平。五个世界在先验臂与独立预算会话间复用，题目不是独立世界重复。三十场全部保留，包括一场来源终检不足。条形表示描述性汇总而非置信区间；均值参考不使用额外实验或隐藏答案。

未知条件的纯度预测均值只有 0.86833，而参考均值为 0.98566，全部 12 个名义 80% 区间均未覆盖目标。这个案例不能用于估计该行为的总体发生率，也不能证明 agent 在心理意义上“知道自己错了”。它支持的更窄结论是：口头承认证据有限，并不自动保证这一限制被正确传播到定量预测区间。

将水相平衡与结晶并置，可以看到两类互补的泛化偏离。在 EQ 的极稀条件下，真实响应离开来源平台，Aligned 预测却继续延伸平台；在结晶纯度中，真实响应在当前测试条件内仍保持高水平，预测却向下偏离。两个体系的底层过程规律均保持固定，变化的是干预条件及其产生的响应。这里的“保持”和“修订”描述预测相对于来源观测的关系，不代表直接测得的内部信念变化。两种结果共同说明，科学泛化需要依据证据判断关系的适用范围；既不能一律复制来源均值，也不能把更积极的外推本身当成进步。

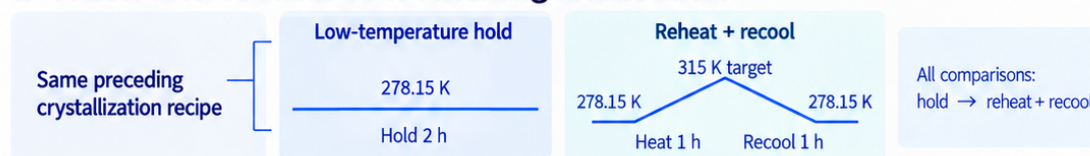
图 4 所示 Aligned 结晶配对案例的后测进一步展示了定量预测与公开反思之间的区别（图 7）。两场原研究会会话均预测复热再冷却将降低细粉比例，而评价器参考给出相反方向。在没有参考反馈的回顾问答中，两者均指出该预测不可靠；24 批次会话还指出，它没有充分利用第 8 批复热未见改善的结果，并提出区分上游条件与冷却作用的实验。该提议未执行，回顾也不能改变已封存预测。因此，这个实例能够定位 agent 公开承认的证据局限，不能据此认定内部错误原因已被识别或预测已得到修复。它展示的是外推边界，不替代上述纯度稳定性证据。

纯化结果为“预测不变性也是科学能力”提供了额外支持。在事后按固定 0.02 分辨率汇总的干预方向分析中，agent 对浓度变化下纯度和回收的小效应判断分别在 14/15 和 15/15 场中正确；对洗涤分段的回收小效应也有 14/15 场正确，但对同一干预的纯度小效应仅 2/15 场判断正确，13/15 场预测了可分辨变化。这些分析依赖预先固定的问题对与事后方向汇总，因此更适合作为支持性结果而非独立结论；但它们与结晶一起说明，自主科学预测必须同时学习“什么会变”和“什么不应明显改变”。

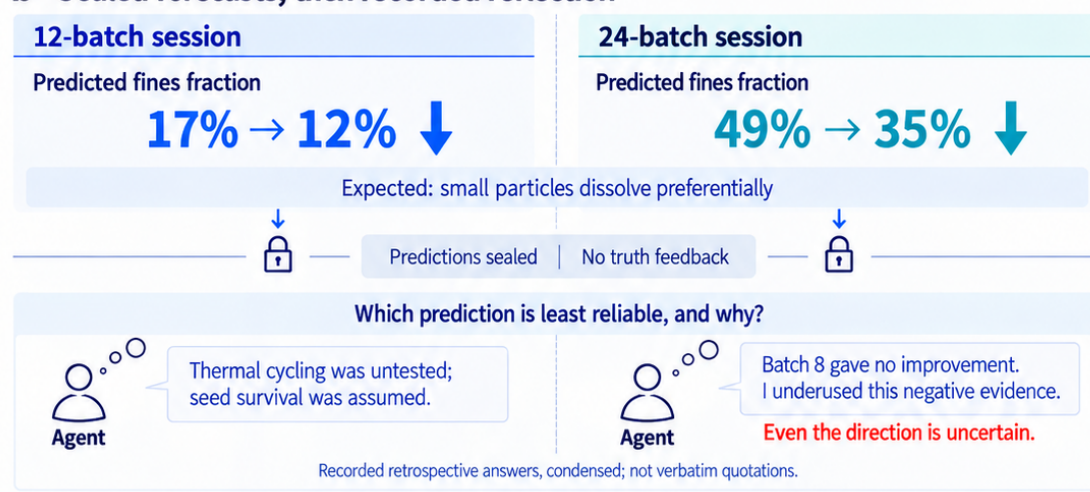
Testing what the experiments support

Blind prediction and reflection in the original research sessions

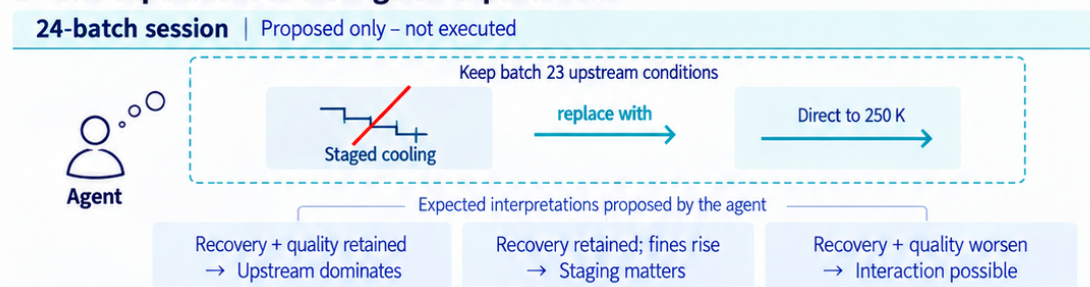
a A new intervention: does reheating reduce fines?



b Sealed forecasts, then recorded reflection



c One experiment to distinguish explanations



d Evaluator reference

Unavailable during prediction and reflection.



Selected case; two independent sessions. Fines: particles <20 micrometres.

Later reflection cannot revise the sealed forecasts.

图7 原研究会话中的盲测、事后反思与未执行提议。a, 共同前序结晶后, 比较低温保持2小时与目标315 K复热1小时后再冷却1小时。b, 两场会话均预测细粉减少; 预测封存后、未获得真值反馈时, 两者指出热循环外推不可靠, 24批次会话还指出未充分利用第8批的负面结果。文字是原回答的压缩转述。c, 24批次会话提出替换第23批冷却方式的判别实验及预期解释, 均未执行。d, 仅针对a中复热题的无噪声参考给出相反方向; 它不是对c中提议的检验, 也没有输入给反思阶段。

这一系列结果最终把问题从“任务成功吗”“MAE 是否下降”推进到关系适用域的判断。一个有用的概念化方式是把真实干预相对于来源关系的偏离与 agent 预测的偏离相比较：真实变化大而 agent 修订不足，对应过度延伸；真实变化小而 agent 修订过强，对应过度修订。本文现有研究并未对全部体系建立统一连续的 revision-calibration 指标，因此我们不把这一框架当作已经完成的全局评分。它更准确地概括了 EQ 与结晶中由数据直接支持的两类互补现象，并为未来系统评价提供了可检验方向。

讨论

ChemWorld 的主要贡献包含两个层面。方法上，它把自主科学研究中通常纠缠在一起的底层规律、实验观测、外部信息和研究决策拆分成可操纵对象，使同一物理世界中的不同研究条件可以被匹配比较，并使完整行动—观测轨迹在研究结束后可重放、可审计。实证上，240 场研究显示，操作交付、预测泛化和与证据相称的不确定性可以明显分离。相分配和反应参数研究中的正面结果尤其重要：它们说明当前 agent 可以在部分任务中通过更多证据或有用信息形成更好的未知条件预测，因此本文不是一个“agent 普遍不会做科学”的失败合集，而是在定位一种选择性的科学泛化问题。

这一问题可以表述为 **scope calibration**：研究者需要判断一个从来源实验中获得的关系，在新的干预下应保留到什么程度、又应在何处被修订。EQ 与结晶恰好给出两个方向相反的证据。如果只采取经验保守策略，结晶纯度可能表现很好，但 EQ 极稀条件会把来源平台错误延伸到新的响应区间；如果只强调机制性变化和积极外推，EQ 中某些问题可能需要这种修订，结晶纯度却会出现不必要偏离。因而不存在一个普适的“更保守”或“更机制化”规则。真正的能力链条更接近：**获得证据 → 形成关系 → 判断适用域 → 在干预下修订 → 给出预测和不确定性**。

研究预算结果进一步提示，决定科学价值的不是样本数量本身，而是新证据能否区分与未知干预相关的竞争关系。PA 中较大 research envelope 与更广材料配对覆盖和更好的预测同时出现；EQ 中 Aligned 来源虽然横跨约 90 倍输入浓度，却仍停留在同一个响应平台。由此，我们认为“input-space coverage”“response-regime coverage”和“discriminating evidence”应当被明确区分。本文的现有轨迹只为这一观点提供行为层面的正反例，还没有给全部 240 场 campaign 统一标注竞争假设与判别实验。因此，更强的因果结论需要在未来固定证据内容、只改变信息或推理条件，或前瞻性地规定能够区分候选模型的实验。

不确定性是关系适用域的另一面。EQ 的最稀问题并非只出现较大点误差，Aligned 同时给出了很窄且几乎完全失效的区间；在结晶个例中，文字报告已经承认证据不足，但数值区间仍未覆盖留出目标。这些现象提示，自主科学中的不确定性不应只反映观测噪声或已选模型内部的参数不确定性，还需要反映“当前关系是否仍适用”的模型或适用域不确定性。不过，本文并未对区间中的 aleatoric 与 epistemic 成分进行显式分解，因此我们把这一点保留为由当前结果支持的评价原则，而非已经识别的误差机制。

本文有几项重要边界。首先，主研究只使用一个 agent 配置，每个研究块有五个复用 world cluster；问题、响应、信息臂和批次并不是相互独立的统计重复，因此我们报告配对方向、效应大小、完整臂均值和影响度检查，而不把查询级观测当作独立样本进行显著性检验。其次，EQ 的极稀分组和结晶回收—纯度联合分析属于保留数据上的事后定位分析，不能被重新包装成预先注册端点。第三，信息臂同时改变 agent 获取哪些证据和如何解释证据，因此当前结果不能因果区分 acquisition 与 inference，也不能证明 prior anchoring、report compression 或任何心理机制。Q 始终保留完整来源上下文，故本文尤其不把 K1 与 Q 的差异解释成报告压缩造成的信息损失。最后，ChemWorld 的平台资格验证建立的是有限声明域中的内部一致性，而不是任意真实化学体系的外部有效性；本文没有湿实验验证，因此结论限于受控计算世界中的自主科学推理行为。

这些限制并不削弱受控世界作为研究仪器的作用，反而界定了下一步实验应当发生在哪里。未来自主科学评价不应只问“agent 是否找到一个好条件”，还应明确测量：它获得了哪些有判别力的证据；它形成的每一个响应关系在什么域内成立；它在新干预下修订了多少；以及它的不确定性是否随证据的可辨识性变化。就这一意义而言，ChemWorld 的目标不是用模拟器替代真实发现，而是建立一个可以被操纵和证伪的中间层，使证据、关系、适用域和预测之间的连接成为可以直接实验研究的对象。

方法

世界构建与执行

一个 ChemWorld 世界由公开实验合同 W_{pub} 与私有过程参数 θ 共同定义。兼容过程组件声明状态变量、参数域、依赖关系和接口，操作与观测模型将这些声明编译为可执行实验环境。agent 只能通过公开操作和仪器与世界交互，不能读取模拟器源码、私有参数、评价器记录或隐藏状态。操作可以消耗物料、时间、操作额度与计算资源；测量本身也可能消耗样品。

世界采用事务式状态更新。每个候选动作首先经过类型、物料、资源与操作合法性检查；合法动作产生候选转移并提交到持久状态，非法或拒绝动作保留已提交物理状态，同时按照合同记录其尝试后果。完整轨迹记录公开观测、执行动作、资源账本和状态转移。精确 replay 指重建环境和已执行操作，而不是重新生成语言模型响应。

平台资格验证

平台资格验证与 agent 科学表现分开。冻结资格覆盖 64 个 task-world 单元、1,786 个边界与分类配方、192 个非法动作探针和 52 个生成组合；生成组合包括 18 个既有参考注册表外拓扑、8 个在已注册拓扑上的新 task-world 身份与 26 个额外覆盖案例。组件和接口检查另包含 32 个模块探针、7 条合法路径和 7 个非法组合。资格验证检查声明域内的可执行性、资源守恒、信息分离、有效 query reference 和 replay 一致性，不作为任意真实化学的验证，也不进入 240 场 agent campaign 的分母。

研究单元与科学任务

当前研究由八个 study block 构成：电化学实体先验 60 场、相分配实体先验 30 场、反应与热过程的参数和结构先验共 60 场、水相平衡的实体/参数/结构先验各 15 场、结晶实体先验 30 场和纯化实体先验 15 场，共 240 场。每个 block 含五个物理世界和三种信息臂。

电化学与反应过程分别设置 discovery 和 optimization commission。电化学、相分配和结晶包含 12 与 24 批次研究资源包，其他块使用 12 批次。参数先验的 P 与纯化体系的 P 是不同概念。电化学改变材料、电位、电流上限与持续时间，optimization 使用公开的平衡效率得分；反应过程改变材料、催化剂负载和热历史；相分配研究材料、体积和过程改变下的两相分配；平衡研究量、体积、浓度、分步添加、解离和沉淀；结晶在纯度不低于 0.80、细粉分数不高于 0.50 且存在颗粒的质量约束下寻求扣除晶种质量后的回收；纯化在纯度不低于 0.80 的约束下寻求原投料回收。结晶与纯化的回收分母不同，不作直接横向合并。

Agent 与信息条件

全部来源研究使用同一 GPT-5.6 Sol 配置与 medium reasoning effort，并在独立会话中运行。一个 block 内不同信息臂共享物理世界、操作空间、资源合同和留出问题。Opaque 保留操作接口和一般任务背景，但不提供被测实例的特定 dossier；Aligned 提供局部、名义匹配的信息，例如 EQ/P 中真实的 effective-pKa 区间，但不提供完整耦合规律；MisIndexed 按 block 的预定规则提供错配或重新索引信息，而非任意随机误导。

agent 可以自主选择条件、重复次数、测量分配与实验分组，并使用受限的公开数值工具。我们不规定固定机理模板、固定方程族或每批实验后的强制反思。批次、底层 operation、模型响应和基于新反馈作出的决策属于不同计数单位。每个 batch budget 同时允许相应数量的额外测量与最终 assay，其他材料、时间、操作和计算额度另行约束。24 批次条件为新的独立研究会话，且具有更大的执行与计算资源，因此预算比较是 research-envelope intervention。

封存评价与独立复测

对于需要交付操作方案的任务，来源研究结束后从已完成批次中封存推荐，并在同一世界中用不同观测种子独立执行，共有 165 次推荐复测。之后，K1 要求 agent 给出自由形式科学说明，包括关系、证据、适用范围与未解决替代解释；Q 要求其在无实验访问和目标反馈的情况下，对 12 个固定新批次条件给出点预测与区间；K2 在 Q 封存后询问证据支持、替代解释、判别实验、目标权衡、未使用证据、不确定性与局限。短版本合并为三个问题，反应与平衡使用七主题版本；EQ/P 另有 15 份结构化参数补充问答。

原始研究上下文在全部阶段均保留，因此 Q 可以同时使用来源历史与 K1，K2 则不能修改已经封存的 Q。当前设计因而不测量“仅通过报告传递知识”的性能，也不把自然语言机理正确性压缩成一个统一分数。

预测目标与评价指标

每个 campaign 对 12 个未知条件进行预测。对于响应 m ，点预测误差定义为 12 个问题上的平均绝对误差：

$$\text{MAE}_{cm} = \frac{1}{12} \sum_{q=1}^{12} \left| \hat{y}_{cqmq} - y_{qm}^{\text{ref}} \right|.$$

区间覆盖率为规定参考目标落入报告区间的比例，区间宽度为平均区间跨度。相分配使用名义 90% 区间，其他当前研究块使用 80%。

不同体系采用不同参考目标。相分配和结晶的点预测针对最终 assay 前的无噪声响应；电化学和纯化使用固定 seeded observation；反应与平衡的点目标为每个条件五次参考观测的均值，而覆盖率针对单次参考观测计算。宏平均误差在指定 block 内对响应等权平均。由于物理量、归一化方式和参考目标不同，本文不构造跨体系 pooled accuracy ranking。

结晶的 public-mean 参考仅使用对应 campaign 的公开来源终检，最近邻参考额外使用公开配方特征。二者在拟合时均不访问隐藏真值。EQ 的 source-mean diagnostic 为独立的事后参考，用于说明在极稀问题中机械保持来源平台并不足够；它不是强系统辨识基线。

匹配比较、探索性分析与失败保留

目标比较按物理世界、信息臂、预算和先验作用层进行匹配；预算比较按世界、信息臂和目标匹配不同独立会话；信息臂比较则在同一 study block 内固定其他条件。五个世界在多个臂、目标和预算下复用，问题、响应与批次存在相关性，因此我们主要报告描述性幅度、配对方向、完整臂均值和影响度分析，而不把所有 query 当成独立重复。

EQ/P 的三个最稀问题由公开浓度事后确定，名义浓度约为 1.33×10^{-4} 、 1.33×10^{-5} 和 1.67×10^{-4} M；按 3:9 加权后可重构原始 12 问宏平均。结晶的回收—纯度联合方向分析同样属于保留数据上的探索性分析。两者都不改变原始未知问题库，也不替换不利结果。回顾性选取的轨迹案例只用于解释 cohort-level pattern，不用于估计发生频率。

全部 240 场计划 campaign 均进入结果。一个结晶来源因溶剂耗尽只完成 11 次最终 assay；一个纯化来源启动 12 个容器后合法丢弃两个，仅完成 10 次最终 assay。二者均标记为 nonconforming 并保留。基础设施恢复保留先前尝试与资源核算。结晶 public baseline 的配方解析器曾遗漏两条 resource-denied transaction；修复后 30/30 baseline 可计算，原有 28 条结果逐项不变，且没有重新调用 agent 或 simulator。所有更正均与 agent 原始预测、独立复测和分母分开记录。

数据与代码可用性

支持本文的来源观测、封存输出、评价器汇总和分析代码均保留在项目 evidence records 中。补充材料报告完整研究矩阵、预测目标、全部响应级信息臂均值、预算比较与探索性分析。用于这一整合研究的公开归档包仍在最终整理中；较早的平台公开版本不应被默认为包含本文后续 agent 研究。

扩展图

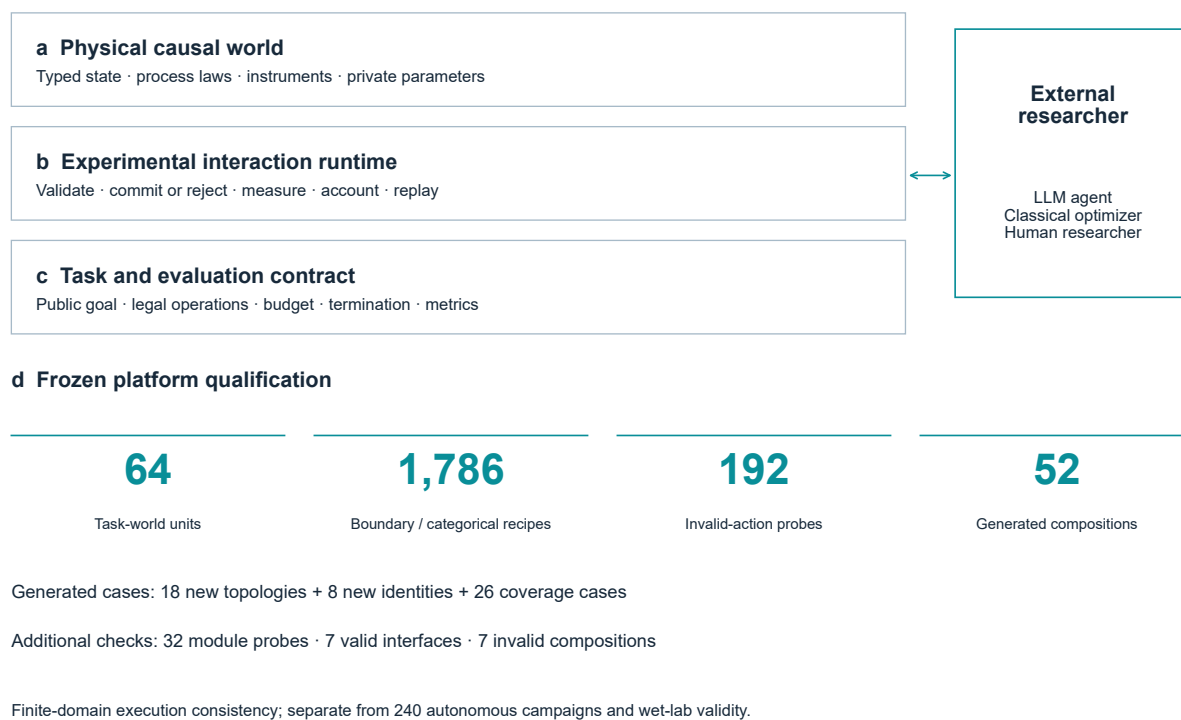


图 S1 可组合基础设施与有限声明域资格。a–c，物理因果层、交互运行时和任务评价合同分离，研究者位于环境之外。d，冻结资格包含 64 个任务—世界单元、1,786 个边界/分类配方、192 个非法动作探针及 52 个生成组合；生成组合进一步分为 18 个新拓扑、8 个新身份和 26 个覆盖案例。另有 32 个模块探针、7 条合法接口和 7 个非法组合。统计建立内部执行一致性，既不属于 240 场 agent 研究，也不等于湿实验有效性。

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补充材料

以下保留既有英文协议说明、全部响应级数据表和探索性分析，便于核对正文分母与证据边界。

A Protocol detail and evidence handling

A.1 Complete study matrix

System / prior locus	Scientific commission	Budgets	Campaigns	Final-assayed batches
EC / E: electrochemistry	Discovery; score optimization	12, 24	60	1,080 / 1,080
PA / E: phase partitioning	Explain phase allocation	12, 24	30	540 / 540
RX / P and S: reaction and thermal processing	Discovery; safety-constrained optimization	12	60	720 / 720
EQ / E: medium identity	Characterize entity-conditioned responses	12	15	180 / 180
EQ / P: bounded equilibrium	Characterize effective responses	12	15	180 / 180
EQ / S: structural equilibrium	Explain response structure	12	15	180 / 180
C / E: crystallization	Deliver crystals under quality constraints	12, 24	30	539 / 540
P / E: purification	Recover product subject to purity	12	15	178 / 180
Total	Six system families		240	3,597 / 3,600

Table A1. Complete selected study scope. Each campaign has one agent realization. One C campaign exhausts its solvent stock and discards its twelfth vessel, completing eleven final assays. One P campaign uses all twelve vessel starts but discards two vessels, leaving ten final assays. All campaigns have sealed mechanism, prediction, and reflection stages; 238 meet the full source-and-assessment specification.

A.2 Prediction targets and system-specific questions

System	Prediction responses	Reference target	Interval level
EC	Six conversion/efficiency responses, including score	Seeded final observation	80%
PA	Organic and aqueous product fractions	Noiseless pre-sampling fractions	90%
RX	Yield, conversion, selectivity, byproduct signal, risk, score	Five-observation mean; coverage over observations	80%
EQ / E, P and S	Normalized pH, dissociation, precipitation signal	Five-observation mean; coverage over observations	80%
C	Net recovery, purity, size index, fines fraction	Noiseless pre-assay responses	80%
P	Purity, original-charge recovery	Seeded final observation	80%

Table A2. Every campaign predicts twelve conditions. PA fractions are complementary, and its two decision questions reuse the conditions. Reference repeats reduce observation noise but do not add agent sessions.

EC's twelve conditions balance positive and negative potential while varying duration, materials, and current cap at a stated loading. Its six responses are selective product yield, electrochemical selectivity, Faradaic efficiency, transport efficiency, energy efficiency, and public score. Current is a magnitude cap, not guaranteed delivered current. PA predicts phase fractions before analytical sampling; sampling losses and complementary fractions must not be counted as separate discoveries.

RX's twelve paths examine time, temperature, catalyst, heating sequence, and quenching. Actual thermal telemetry matters: requested boundary temperature need not be realized reaction temperature. Five reference observations per condition are withheld from agents, with all source assessments sealed before reference-truth release. RX's 600 reference executions are separate from 720 source batches and 60 retests.

Each EQ block uses twelve questions and five observations per question, giving 300 reference executions. EQ/E crosses three medium identities with three concentrations and a matched-concentration scale control, assessing entity contrast and concentration and scale transfer under a fixed common topology. Bounded EQ/P examines effective aqueous relationships; canonical EQ/S supports response-shape analysis without requiring a named hidden family. An instrument's equilibrium diagnostic is not the agent's subjective confidence. The bounded study's effective-parameter supplement is additional to K1/Q/K2 and does not confer discovery credit for a parameter already supplied as a treatment.

C's twelve conditions form six pairs: material, seed dose, cooling history, prior thermal history, continued growth, and upstream loading. Its size response is $\min(d_{50}/250\text{ }\mu\text{m}, 1)$, a bounded number-weighted index rather than a diameter in micrometres. Fines are particles below 20 micrometres. Quality requires purity and fines criteria jointly, with particles present; recovery of 0.10 is an initial feasibility target, not an optimization ceiling. P's six paired interventions concern multistage recovery, including wash staging. Its original-charge denominator differs from C's crystallization-stage recovery.

RX and EQ coverage uses individual reference observations while point targets use means. EC and P use single seeded observations. PA and C prediction coverage concerns noiseless pre-sampling responses; C recommendation retests use observed assays. Apparent retest threshold crossings can therefore reflect noise. Macro errors weight a system's responses equally; normalized scales do not make responses physically or scientifically interchangeable.

A.3 Language, numerical tools, and stage budgets

Source briefs and requested reports are English. EC/PA and C/P assessment instructions are English; RX and EQ K1/Q/K2 instructions include Chinese text requesting English reports. Within-block arms have the same language contract. Seven-theme K2 reviews cover treatment claims, influential experiments, competing explanations, one proposed discriminating experiment, goal-related tradeoffs, unused evidence and uncertainty, and limitations of the recommendation or scientific account. Three-question versions combine these themes. The reviews occur after source experiments and sealed predictions; their length does not provide additional source experiments.

The earlier EC/PA block retains its saved eight-call diagnostic-calculator hard stop; later blocks disclose 128-attempt assessment allowances. An old EC parameter/structural supplement encountered the smaller limit and remains an execution failure outside the primary E cohort. The repair is not retroactively attributed to older sessions. Public calculation uses agent-accessible information, not hidden targets. Differences in numerical budgets and deadlines are execution contracts, not scientific findings.

A.4 Completion, recovery, and exclusions

The primary denominator is 240 scheduled campaigns, not provider attempts. All 720 K1/Q/K2 stages and 165 planned recommendation retests in EC, RX, C, and P are complete, plus fifteen EQ supplements. Final-assayed source batches total 3,597 of 3,600 planned. C's second-world 12-batch Opaque source has eleven final assays. P's fifth-world Opaque source starts twelve vessels, discards two, and completes ten final assays. Both remain in outcome tables with conformance indicators.

EC/PA has 68 first-attempt and 22 infrastructure-recovered campaigns. Recovery after transport or host failures retains earlier attempts and consumed resources. Missing assessment stages resume from saved context when the source is intact; fresh source replacements are recorded as such. Repeated question delivery and failed-attempt computation are not silently counted as an original clean attempt. EC goal sensitivity is reported in the main text. First-attempt-only EC-discovery, EC-optimization, and PA budget comparisons improve prediction in four of five, eight of eleven, and eight of ten pairs, respectively; these reduced subsets have uneven coverage.

The current P matrix follows a correction of prepartition inventory handling. Affected qualification and the entire agent block were restarted under the corrected contract; earlier apparently successful cells were not selected into the new matrix. All current source and recommendation trajectories pass inventory, resource, and replay checks. Its two lawful discards are a source-assay shortfall, not an inventory failure. The fixed purity-direction rule treats changes of 0.02 or less as small; a numerical tolerance of 10^{-12} corrects floating-point equality handling. Six direction classifications changed after this correction, with originals preserved. Predictions, point errors, intervals, retests, and denominators were unchanged.

Historical single-world EC parameter/structural supplements, the closed-set EQ/S pilot, early C/P development runs, qualification recipes, references, replays, and interrupted attempts do not enlarge the denominator. Their records remain separate. Frozen platform qualification is also a separate evidence programme. These counts describe execution and analysis scope; they are not 240 independent chemistry replications.

A.5 Baseline correction and trace interpretation

The C empirical references use the same source final observations for all three arms. A recipe parser originally rejected resource-denial transaction records in two sources. The correction preserves those attempts as provenance and excludes them from committed physical recipe features. Recomputing all thirty baselines restores the two missing evaluations and exactly reproduces the twenty-eight previously available results. Original trajectories, truth, model answers, and error records are retained; the correction uses no new simulator or provider calls. Unknown transaction statuses remain errors. This is an evaluator repair, not a change to a completed experiment.

The EC illustrative trace is the second-world, 12-batch Opaque EC discovery campaign; its sparse-positive-coverage counterexamples are first-world, 24-batch Aligned and MisIndexed discovery campaigns. They are retrospectively selected examples, not a coded prevalence estimate. The additional C case is the fifth-world, 12-batch MisIndexed campaign, chosen as the largest purity MAE in the 12-batch cohort. Its extreme error illustrates a failure; it is not a representative sampling rule.

A systematic analysis should attach each mechanistic claim to its experimental support, contradictory observations, stated domain, and corresponding predictions. It should distinguish an unperformed discriminating experiment, acquired evidence omitted from an account, justified unresolved alternatives, and report-prediction inconsistency. Evaluator knowledge helps identify discriminating interventions; a claim is not wrong merely for using an equivalent representation. Independent review can assist this analysis without replacing measured predictions and retests. No completed cohort-wide mechanism score or causal compression experiment is implied.

A.6 Full prior overview

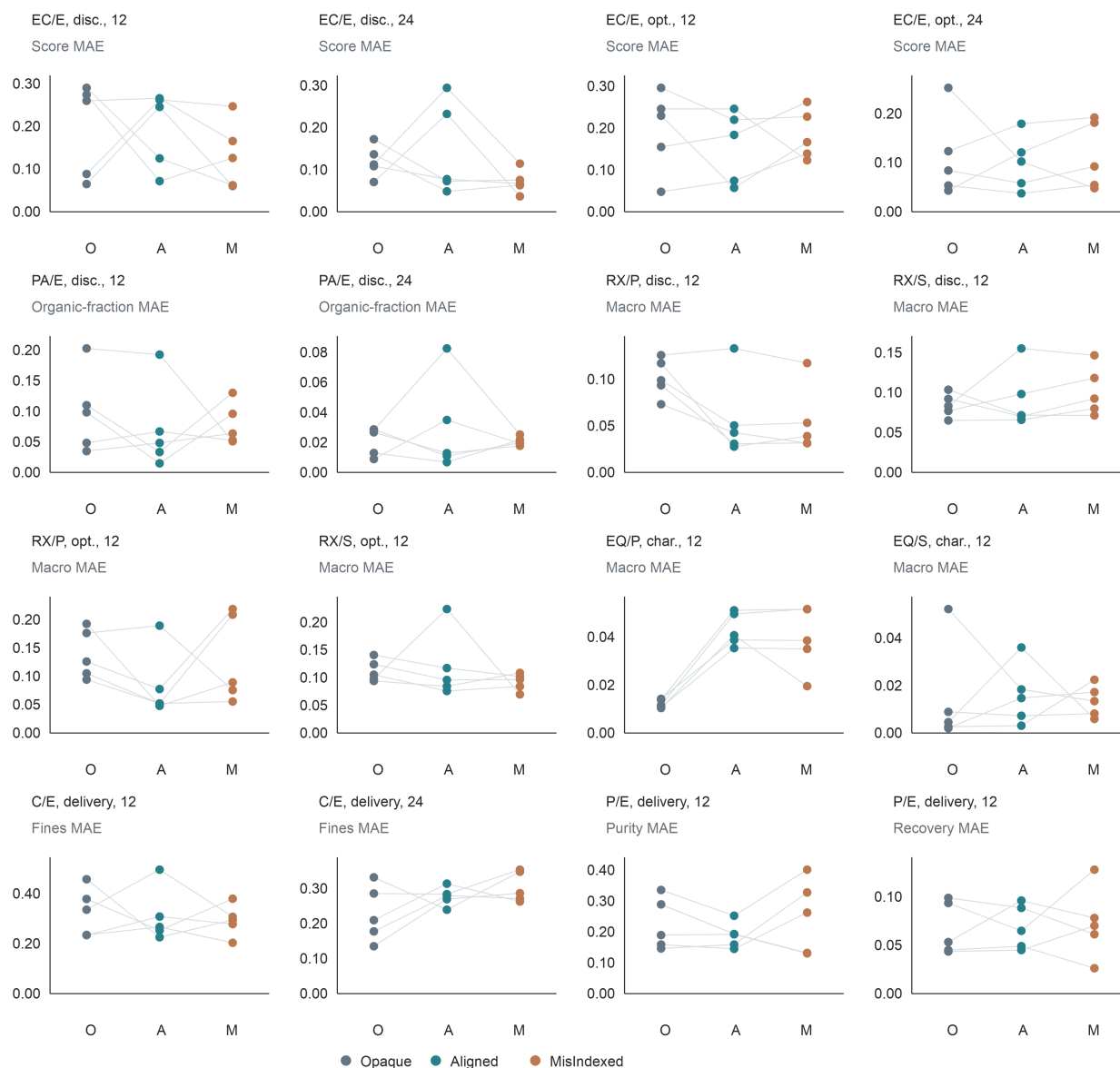
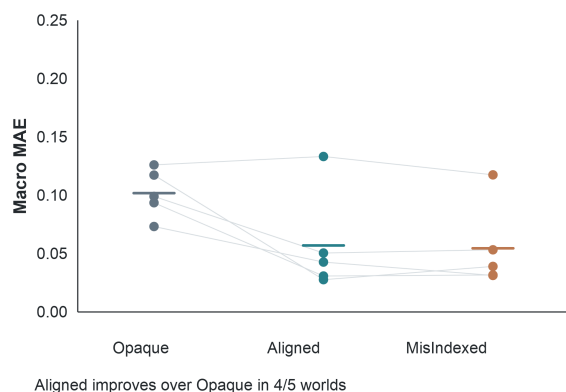


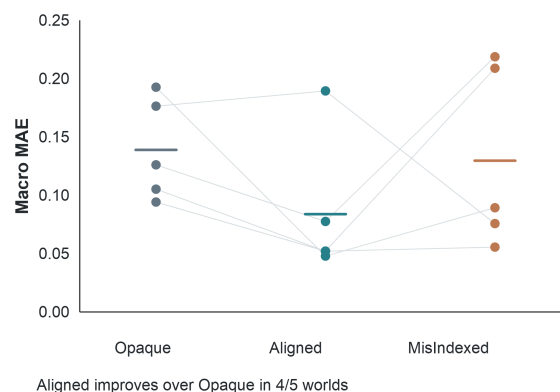
图 S2 Prior information has no common ordering. Each panel retains five worlds and three information conditions: Opaque (O), Aligned (A) and MisIndexed (M). Grey lines connect the same world across arms; colored points show individual campaigns, including source-assay shortfalls. RX and EQ use within-system macro errors; other panels name the response. Scales differ. The complete response tables include the equilibrium entity-prior study and all 240 campaigns.

A.7 Prior effects by system and regime

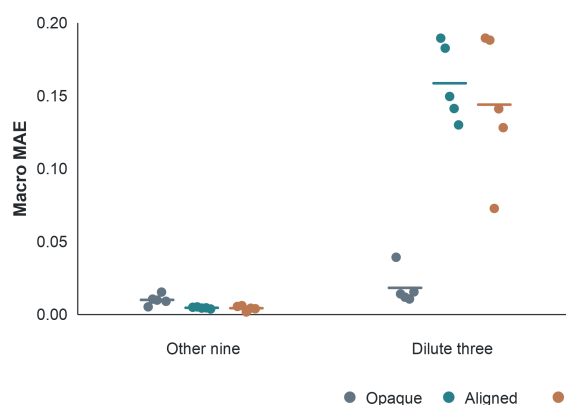
a Reaction processing: discovery



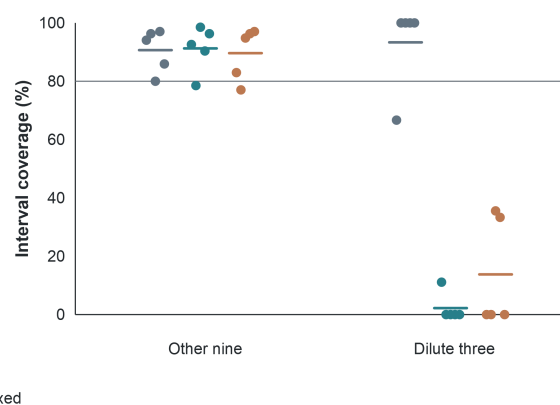
b Reaction processing: optimization



c Equilibrium: parameter prior



d Equilibrium: parameter prior



Points: five worlds per arm; short lines: means. EQ grouping is exploratory.

图 S3 Prior information depends on the system and intervention regime. a,b, Reaction parameter-prior macro MAE under discovery and optimization; lines connect the same world across arms. c,d, Equilibrium macro MAE and interval coverage on the other nine and three most dilute queries. Points retain five worlds per arm; short lines show means. The horizontal reference marks nominal 80% coverage. The exploratory grouping does not classify all other nine queries as interpolation.

A.8 Complete budget comparison

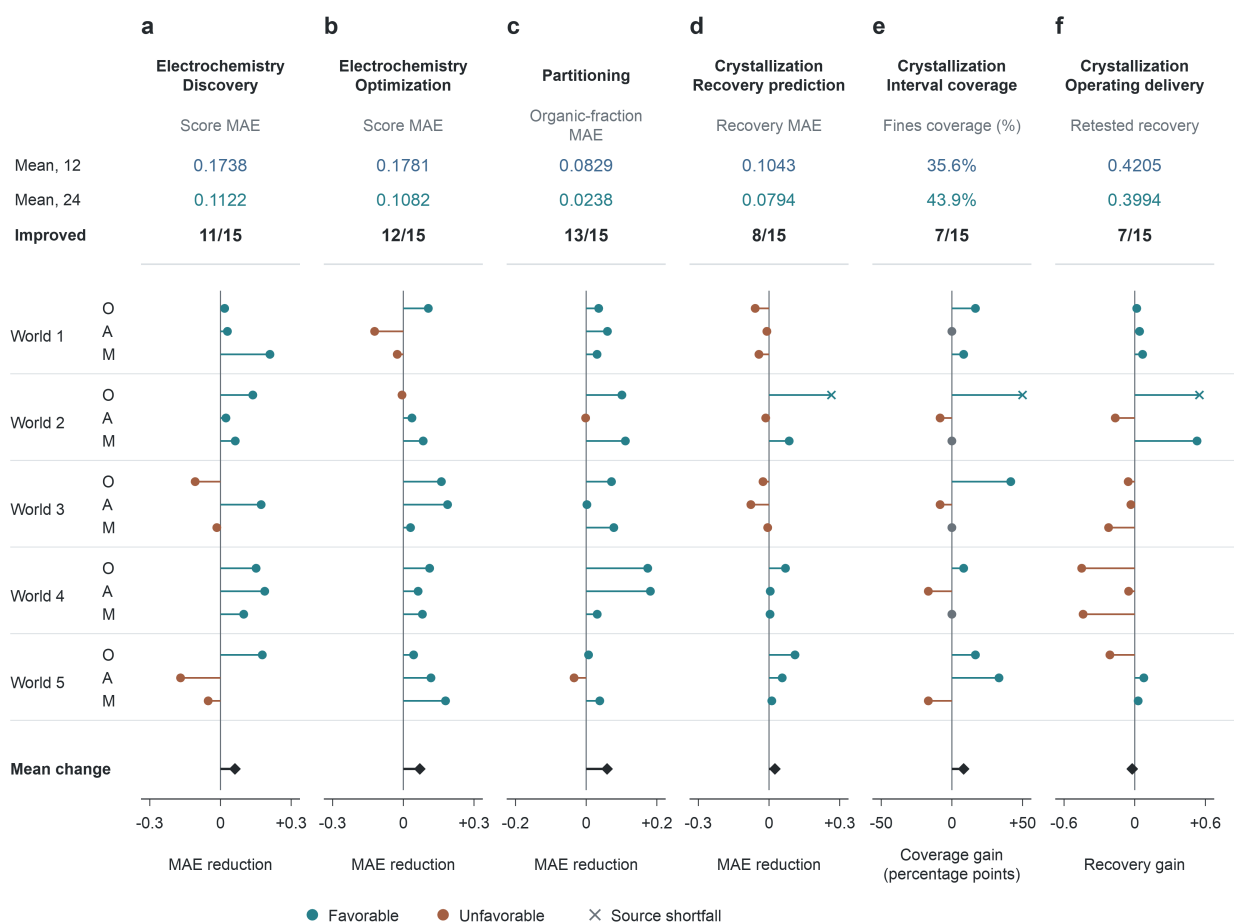


图 S4 Complete budget effects across six readouts. Each column retains fifteen pairs of independent sessions. Headers show campaign means and improved counts; rows show paired changes. Positive changes favour the larger research envelope. Panels a-d show MAE at 12 minus MAE at 24; e shows fines coverage at 24 minus coverage at 12 in percentage points; f shows the corresponding gain in retested recovery. All fines coverage values remain below nominal 80%. Diamonds show mean changes; crosses retain source-assay shortfalls. O, A and M denote Opaque, Aligned and MisIndexed. World labels are system-specific. Scales differ except in a-b. The crystallization panels reuse the same campaigns; larger budgets also increase computational allowances.

B Complete metric-specific arm means

Each entry is mean absolute error / empirical interval coverage (%), averaged over five world sessions. All scheduled sessions are retained, including the two source-budget shortfalls. PA uses nominal 90% intervals; all other studies use 80%. Metric scales and reference targets differ across systems. The machine-readable table includes each campaign's interval width and protocol status.

B.1 EC, E prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.2983 / 55.0	0.2740 / 65.0	0.1272 / 88.3
Energy efficiency	0.3098 / 33.3	0.2595 / 51.7	0.1582 / 63.3
Faradaic efficiency	0.2580 / 46.7	0.2259 / 60.0	0.1356 / 73.3
Public score	0.1953 / 51.7	0.1939 / 60.0	0.1321 / 71.7
Selective product yield	0.0731 / 60.0	0.0817 / 38.3	0.0598 / 56.7
Transport efficiency	0.2602 / 45.0	0.2184 / 66.7	0.1328 / 76.7

B.2 EC, E prior: optimization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.2908 / 63.3	0.2378 / 66.7	0.2661 / 73.3
Energy efficiency	0.2794 / 43.3	0.2541 / 31.7	0.2440 / 55.0
Faradaic efficiency	0.2506 / 53.3	0.2532 / 31.7	0.2315 / 56.7
Public score	0.1947 / 55.0	0.1561 / 58.3	0.1835 / 53.3
Selective product yield	0.0758 / 43.3	0.0623 / 53.3	0.0762 / 50.0
Transport efficiency	0.2478 / 58.3	0.2053 / 51.7	0.2217 / 60.0

B.3 PA, E prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Aqueous fraction	0.0989 / 76.7	0.0712 / 80.0	0.0786 / 78.3
Organic fraction	0.0989 / 76.7	0.0712 / 80.0	0.0786 / 78.3

B.4 EC, E prior: discovery, 24 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.1902 / 68.3	0.1268 / 80.0	0.0428 / 93.3
Energy efficiency	0.1581 / 61.7	0.1651 / 68.3	0.0945 / 80.0
Faradaic efficiency	0.1458 / 65.0	0.1505 / 66.7	0.0877 / 78.3
Public score	0.1201 / 63.3	0.1452 / 73.3	0.0713 / 78.3
Selective product yield	0.0726 / 53.3	0.0573 / 70.0	0.0374 / 75.0
Transport efficiency	0.1495 / 65.0	0.1501 / 66.7	0.0846 / 78.3

B.5 EC, E prior: optimization, 24 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.1403 / 85.0	0.1252 / 85.0	0.0864 / 90.0
Energy efficiency	0.1977 / 48.3	0.1307 / 60.0	0.1779 / 65.0
Faradaic efficiency	0.1502 / 76.7	0.1098 / 78.3	0.1515 / 71.7
Public score	0.1113 / 75.0	0.0996 / 71.7	0.1136 / 75.0
Selective product yield	0.0515 / 75.0	0.0498 / 56.7	0.0587 / 58.3
Transport efficiency	0.1420 / 80.0	0.1257 / 73.3	0.1570 / 68.3

B.6 PA, E prior: discovery, 24 batches

Readout	Opaque	Aligned	MisIndexed
Aqueous fraction	0.0251 / 100.0	0.0296 / 96.7	0.0207 / 100.0
Organic fraction	0.0211 / 100.0	0.0296 / 96.7	0.0207 / 100.0

B.7 RX, P prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.0800 / 88.7	0.0791 / 73.3	0.0710 / 81.0
Conversion	0.0426 / 72.7	0.0321 / 70.0	0.0306 / 73.3
Safety risk	0.1031 / 58.3	0.0282 / 85.0	0.0370 / 81.7
Public score	0.1195 / 48.7	0.0501 / 83.3	0.0476 / 90.3
Selectivity	0.1198 / 58.0	0.0818 / 69.3	0.0714 / 77.3
Yield	0.1461 / 35.7	0.0708 / 76.7	0.0698 / 78.3

B.8 RX, P prior: optimization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.1509 / 57.7	0.0805 / 73.0	0.1271 / 60.3
Conversion	0.0474 / 85.3	0.0319 / 82.7	0.1960 / 53.0
Safety risk	0.0984 / 56.7	0.1106 / 56.7	0.0811 / 68.3
Public score	0.1341 / 30.0	0.0818 / 73.0	0.0879 / 68.3
Selectivity	0.1900 / 31.0	0.0923 / 69.7	0.1360 / 64.7
Yield	0.2128 / 19.3	0.1060 / 65.7	0.1499 / 68.7

B.9 RX, S prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.1219 / 78.7	0.1212 / 75.3	0.1279 / 64.7
Conversion	0.0217 / 97.7	0.0266 / 95.3	0.0200 / 98.0
Safety risk	0.0664 / 80.0	0.0717 / 66.7	0.1276 / 43.3
Public score	0.0504 / 89.7	0.0816 / 61.3	0.0665 / 78.0
Selectivity	0.1280 / 72.7	0.1319 / 70.7	0.1400 / 55.0
Yield	0.1157 / 81.7	0.1196 / 75.7	0.1275 / 61.0

B.10 RX, S prior: optimization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.1376 / 63.7	0.1490 / 67.7	0.1188 / 66.3
Conversion	0.0370 / 88.7	0.0505 / 82.7	0.0425 / 88.3
Safety risk	0.1000 / 45.0	0.1255 / 68.3	0.1133 / 45.0
Public score	0.0932 / 56.0	0.0917 / 58.0	0.0637 / 81.0
Selectivity	0.1574 / 56.7	0.1570 / 60.3	0.1175 / 62.3
Yield	0.1485 / 63.3	0.1422 / 66.3	0.0975 / 73.0

B.11 EQ, P prior: characterization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Dissociation fraction	0.0129 / 96.3	0.0759 / 68.7	0.0646 / 74.7
Normalized pH	0.0031 / 94.3	0.0160 / 65.0	0.0150 / 68.7
Precipitation signal	0.0205 / 83.3	0.0376 / 73.3	0.0383 / 68.7

B.12 EQ, S prior: characterization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Dissociation fraction	0.0292 / 91.7	0.0235 / 76.3	0.0122 / 87.7
Normalized pH	0.0097 / 87.3	0.0200 / 75.7	0.0203 / 77.7
Precipitation signal	0.0033 / 95.3	0.0042 / 95.7	0.0078 / 87.3

B.13 EQ, E prior: characterization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Dissociation fraction	0.0128 / 83.7	0.0107 / 85.0	0.0134 / 79.0
Normalized pH	0.0233 / 80.7	0.0278 / 79.0	0.0202 / 82.0
Precipitation signal	0.0084 / 89.3	0.0116 / 92.0	0.0103 / 88.3

B.14 C, E prior: delivery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Fines fraction	0.3279 / 25.0	0.3096 / 38.3	0.2925 / 43.3
Crystal purity	0.0095 / 98.3	0.0497 / 60.0	0.0917 / 23.3
Particle-size index	0.0777 / 55.0	0.0624 / 58.3	0.0725 / 58.3
Net crystal recovery	0.1632 / 48.3	0.0606 / 83.3	0.0892 / 71.7

B.15 C, E prior: delivery, 24 batches

Readout	Opaque	Aligned	MisIndexed
Fines fraction	0.2281 / 51.7	0.2774 / 38.3	0.3046 / 41.7
Crystal purity	0.0197 / 95.0	0.0398 / 73.3	0.0508 / 55.0
Particle-size index	0.0566 / 71.7	0.0440 / 70.0	0.1023 / 48.3
Net crystal recovery	0.0911 / 61.7	0.0684 / 80.0	0.0786 / 73.3

B.16 P, E prior: delivery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Purity	0.2240 / 51.7	0.1883 / 53.3	0.2508 / 50.0
Recovery	0.0668 / 56.7	0.0687 / 51.7	0.0727 / 53.3

C Exploratory analyses of response regimes

These analyses use the retained source campaigns and their original predictions. They do not define new experimental arms or alter the withheld-condition bank. The concentration grouping, joint crystallization contrast and illustrative traces were selected after the experiments, and are descriptive analyses of reused worlds.

C.1 Equilibrium: a reversal across public concentration regimes

The three most dilute recipes have nominal concentrations approximately 1.33×10^{-4} , 1.33×10^{-5} and 1.67×10^{-4} M. Each campaign has three normalized responses for each recipe. Point errors use five-observation means; interval coverage uses the individual reference observations. Group means average the corresponding responses and queries and then the five world campaigns. Weighting the two groups 3:9 reconstructs every original twelve-question campaign macro MAE. The other nine queries include boundary conditions; this split is not a predeclared interpolation-versus-extrapolation test.

Query group / arm	Macro MAE	Coverage (%)	Interval width	Source-mean MAE
Other nine / Opaque	0.01008	90.7	0.04706	0.00553
Other nine / Aligned	0.00465	91.3	0.03340	0.00518
Other nine / MisIndexed	0.00439	89.6	0.02798	0.00494
Three most dilute / Opaque	0.01833	93.3	0.15676	0.16961
Three most dilute / Aligned	0.15862	2.2	0.08084	0.16833
Three most dilute / MisIndexed	0.14397	13.8	0.09633	0.16836

Table C1. EQ/P subgroup results. Each row averages five worlds; rows reuse those worlds. The source-mean predictor is an additional post hoc diagnostic and obtains no new experiments. Its poor dilute-query performance shows why retaining the plateau is insufficient; its usefulness on the other nine depends on their response distribution.

Aligned improves on the other nine and worsens on the dilute three relative to Opaque in all five worlds. All fifteen source campaigns remain above 0.018 M. In the selected second-world comparison, Aligned spans about ninety-fold source concentration variation and Opaque about fourfold, but both remain in the observed plateau. At the most dilute question, reference dissociation is 0.65463; Opaque predicts 0.629 [0.45, 0.80] and Aligned 0.0804 [0.053, 0.108]. The Opaque prediction response develops an unsaturated weak-acid extrapolation beyond its sealed report. This reasoning must not be credited retrospectively to K1 or to an earlier source-stage hypothesis.

C.2 Crystallization: joint response changes under the same treatment

Budget / arm	Recovery MAE	Purity MAE	Recovery improves and purity worsens vs Opaque
12 / Opaque	0.16322	0.00952	Reference
12 / Aligned	0.06055	0.04969	5 / 5 worlds
12 / MisIndexed	0.08923	0.09166	5 / 5 worlds
24 / Opaque	0.09107	0.01975	Reference
24 / Aligned	0.06839	0.03983	2 / 5 worlds
24 / MisIndexed	0.07863	0.05080	3 / 5 worlds

Table C2. Joint directional comparisons use both response errors from the same world, arm and budget. At twelve batches, each joint direction also holds in all four pairs whose two campaigns conform to the source-assay contract. The direction weakens at 24 batches. A treatment can therefore be helpful on one response and harmful on another without implying a budget-invariant tradeoff.

C.3 Evidence status and interpretation

The same-source C references use only each campaign’s final assays and public recipe features. They exclude hidden truth during fitting but do not reproduce the agent’s complete intermediate measurements, prior or computational policy. Low held-out purity variation makes the mean a particularly strong reference. Neither its success nor the agent’s recovery advantage is a complete mechanism identification result.

Selected EC and C traces support detailed examples rather than population frequencies; the bounded EQ review below covers all fifteen campaigns in that block. The current evidence does not causally identify prior anchoring, report compression, acquisition-versus-inference mediation or a general preference for familiar chemistry. Full-history Q also prevents interpreting K1 as the sole information channel. Experimental selection and later interpretation are both part of the original autonomous process. Their separate causal contributions are not required to describe the observed whole-process contrasts and have not been identified here.

C.4 Original equilibrium campaigns: predictions and research chronology

We reviewed all fifteen retained parameter-prior equilibrium campaigns, comprising 180 batches, 945 operations and 45 K1/Q/K2 stages. Each row below is an original autonomous research session, not an independent reader of its records. The selected question uses 1 micromole in 75 mL; the same question and original reference targets apply to all three arms in a world. The reference column is the five-observation mean.

World / arm	Public Q explanation	Dissociation [80% interval]	Reference
1 / Opaque	Piecewise weak acid	0.6990 [0.250, 0.900]	0.7160
1 / Aligned	Plateau / weak trend	0.1040 [0.044, 0.225]	0.7160
1 / MisIndexed	Plateau / weak trend	0.1092 [0.0772, 0.1412]	0.7160
2 / Opaque	Piecewise weak acid	0.6290 [0.450, 0.800]	0.6546
2 / Aligned	Plateau / weak trend	0.0804 [0.053, 0.108]	0.6546
2 / MisIndexed	Plateau / weak trend	0.0806 [0.0556, 0.1056]	0.6546
3 / Opaque	Piecewise weak acid	0.5630 [0.430, 0.700]	0.5797
3 / Aligned	Plateau / weak trend	0.1100 [0.045, 0.190]	0.5797
3 / MisIndexed	Mixed weak acid / plateau	0.4600 [0.100, 0.750]	0.5797
4 / Opaque	Qualitative departure	0.5200 [0.150, 0.880]	0.5173
4 / Aligned	Plateau / weak trend	0.0724 [0.040, 0.105]	0.5173
4 / MisIndexed	Plateau / weak trend	0.0962 [0.040, 0.450]	0.5173
5 / Opaque	Piecewise weak acid	0.4130 [0.200, 0.700]	0.4480
5 / Aligned	Plateau / weak trend	0.0790 [0.035, 0.130]	0.4480
5 / MisIndexed	Plateau / weak trend	0.0681 [0.040, 0.096]	0.4480

Table C3. Public extrapolation accounts and original same-context predictions. Categories are a single-author retrospective reading of public explanations, not internal reasoning traces or independent causal labels. Four Opaque accounts use piecewise weak-acid reasoning; the fifth permits a qualitative departure. All five Aligned accounts extend a plateau or weak trend. The third-world MisIndexed mixed account is a counterexample to treating all dossier conditions alike.

The first-world Opaque K1 already estimates an effective acid constant near 2.2×10^{-5} and an active-pool cap near 0.004 M, while retaining competing interpretations. Its Q applies that hypothesis at low concentration. In the second world, Opaque K1 primarily supports the observed plateau and limits extrapolation; the unsaturated weak-acid calculation is first explicit in Q. The aligned K1 similarly limits its account to the sampled region, but its Q interval still misses the dilute response by a large margin. These differences separate what was stated before prediction from what became explicit during it.

All fifteen source concentration ranges lie above the three dilute test recipes. The exported per-operation decision-reason fields are unavailable for all 945 operations. Actions and observations can therefore be reconstructed, but a contemporaneous sequence of belief revisions cannot be inferred from these fields. K1, Q and K2 remain distinct public records rather than substitutes for that sequence.

In K2, eleven of fifteen campaigns propose a trace-load experiment: five Opaque, three Aligned and three MisIndexed. The other choices are equal-concentration scale-up, dilution near the supplied relation, comparison of instruments at the same state and repetition of an anomalous batch. K2 follows exposure to Q conditions without feedback on their reference outcomes. None of these proposals was executed; they are retrospective choices, not source-stage plans or demonstrated repairs.

D Selected autonomous research histories

Figure 4 and Supplementary Fig. S5 follow the same retrospectively selected Aligned crystallization pair. Each budget is an independent session. The case documents actions, observations and later public accounts; it does not estimate the cohort-wide effect of additional resources.

D.1 Experimental paths, recommendations and retests

A retrospectively selected Aligned crystallization pair illustrates how independent sessions can follow different experimental paths (Fig. 4). The 12-batch session first obtains a quality-feasible product in batch 1 and later lowers the cooling endpoint from 260 to 250 K. The 24-batch session first becomes feasible in batch 20, after changing heating and introducing staged cooling. Its selected procedure has higher independently retested recovery and lower fines than the smaller session's selection. This example illustrates recorded actions and outcomes, not a population budget effect. Contemporaneous decision reasons were not recorded; the illustration therefore shows recorded changes without reconstructing immediate thoughts.

The 12-batch session first meets the quality constraints at batch 1, with recovery 34.7%, and reaches its best feasible observed recovery of 51.2% at batch 10. The 24-batch session first becomes feasible at batch 20 and reaches 57.0% at batch 23. Quality feasibility and subsequent optimization are distinct milestones. All 36 original batch results remain in the source data, including the infeasible trials.

Quench in this environment stops reaction chemistry while subsequent cooling and crystal growth remain possible. The two sessions differ in materials and multiple process conditions; their comparison does not isolate a quench effect. Heating and cooling values are requested targets, not measured temperature trajectories. The K1 summaries were written after the source experiments and are distinct from contemporaneous decision logs.

D.2 Sealed predictions and subsequent reflection

Figure S5 follows the same sessions through sealed prediction and subsequent reflection; the proposed experiment was not executed.

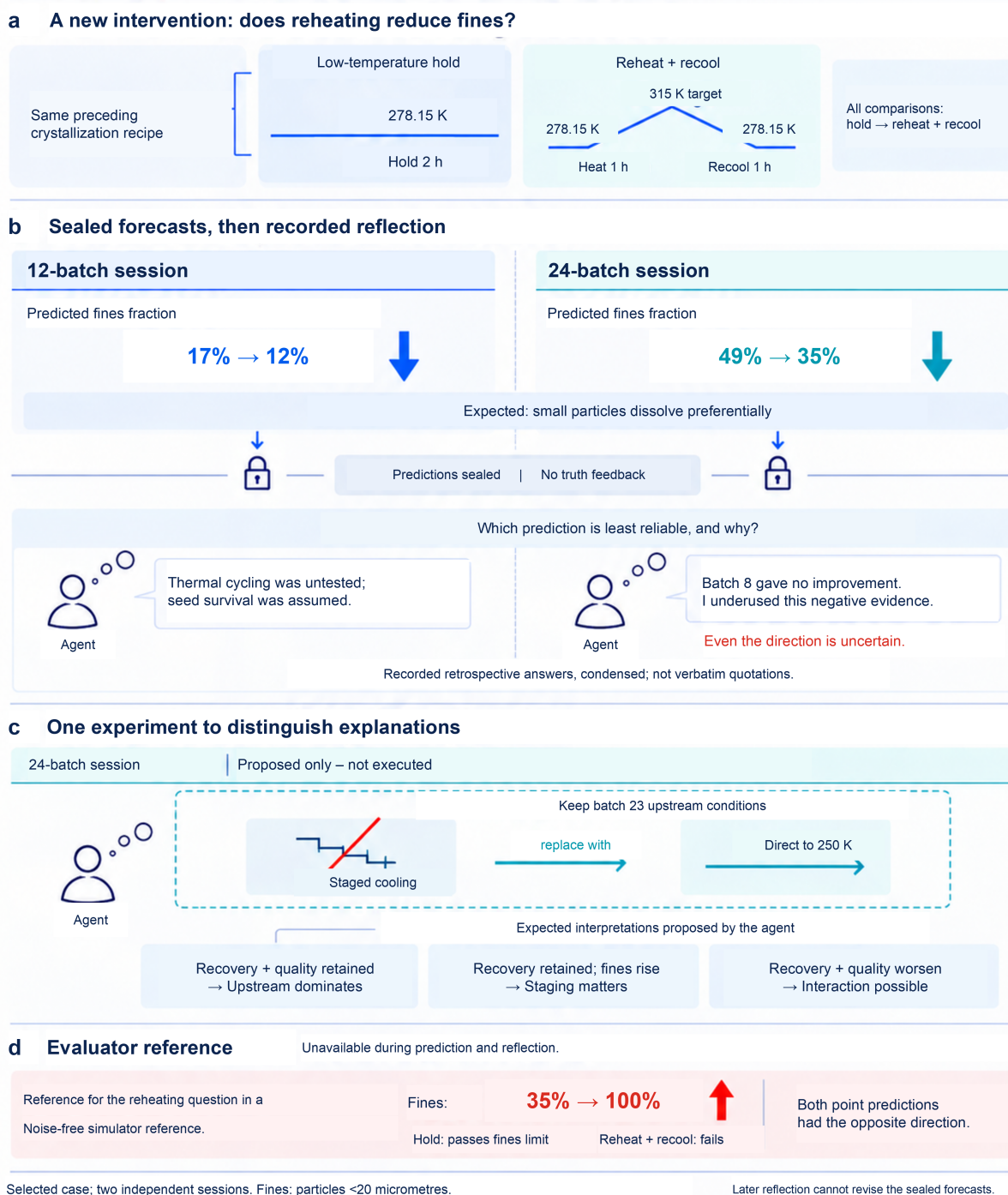


图 S5 Sealed predictions and subsequent reflection in the same selected crystallization pair. a, The withheld comparison replaces a two-hour hold at 278.15 K with heating towards 315 K for one hour and recooling for one hour after a common preceding recipe. Thermal profiles are schematic. b, Both original sessions predict lower fines. The condensed public K2 answers follow sealed Q predictions without reference feedback. c, The 24-batch session proposes changing only the cooling procedure of its selected batch; possible outcomes would distinguish upstream, cooling and interaction explanations. This proposal was not executed. d, The evaluator's reference concerns the thermal intervention in a, not the proposed experiment in c, and was unavailable during Q and K2.

E Supporting outcomes and examples

E.1 Quality constraints and response-specific references

Purification gives a related distinction between recovery and acceptable delivery. Aligned recommendations meet the purity threshold in three of five worlds, compared with none for Opaque or MisIndexed. Yet mean recovered fractions are 0.04455, 0.12373 and 0.32105 for Aligned, Opaque and MisIndexed, respectively. Recovering more material therefore does not imply satisfying the commission. A single retest near the purity threshold is also not a precise estimate of a procedure's reliability. These system-specific outcomes cannot be represented faithfully by a universal yield score (Fig. S6a).

Purification provides another response-specific example: fourteen of fifteen campaigns correctly judge a small concentration effect on purity, but only two correctly judge the small purity effect of splitting a wash. Recognizing approximate invariance is itself a predictive achievement, and depends on the intervention.

In crystallization, agent size forecasts have lower MAE than the source mean in four of thirty campaigns and fines forecasts in twenty-one (Fig. S6b,c). Together with the recovery and purity contrasts in the main text, these results retain the complete response-specific comparison.

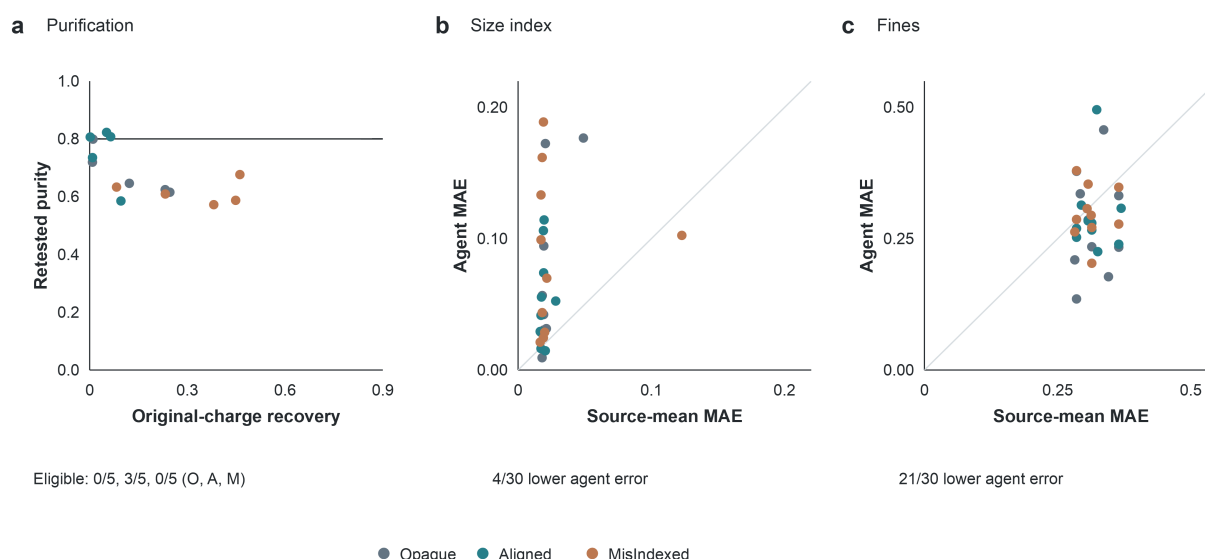


图 S6 Additional operational and predictive outcomes. a, All fifteen purification recommendations, showing original-charge recovery and independently retested purity. The line marks the purity threshold of 0.80; eligible counts are 0/5, 3/5 and 0/5 for Opaque, Aligned and MisIndexed. b,c, Size and fines MAE for all thirty crystallization campaigns relative to each campaign's public observation mean. Points above the diagonal favour the mean; the campaign with a source-assay shortfall remains included. These panels use different response scales.

E.2 Additional selected examples

In a selected Aligned phase-partitioning campaign pair, the larger budget covers all sixteen material pairings and then allocates eight experiments to process variation; the smaller campaign covers ten pairings. This is a plausible route to improvement, not a cohort-wide causal explanation. Two Aligned world pairs nevertheless worsen.

Uncertainty is part of the same applicability problem. The poor dilute-equilibrium coverage shows that a wrong extrapolation can also be overconfident. In a retrospectively selected MisIndexed crystallization campaign, the mechanism report explicitly acknowledges that only one solvent was tested. Nevertheless, predicted purity averages 0.86833 against a reference mean of 0.98566, and all twelve nominal 80% intervals miss. A verbal statement of uncertainty does not by itself ensure calibrated quantitative prediction.

E.3 Complete crystallization baseline and interval comparisons

Figure 6b reports how often each response is predicted more accurately than the public observation mean. Table E1 supplies the corresponding error magnitudes and the public nearest-neighbour comparison. Each baseline is fitted separately to the same campaign’s public final assays; neither receives withheld outcomes or additional experiments. Nearest-neighbour prediction also uses public recipe features. These are empirical references rather than strong system-identification baselines.

Table E1. Crystallization prediction errors and baseline comparisons.

Response	Agent MAE	Mean MAE	Nearest MAE	Wins: mean	Wins: nearest
Recovery	0.09185	0.17135	0.17308	26/30	26/30
Purity	0.04354	0.00482	0.00905	0/30	4/30
Size index	0.06923	0.02345	0.02708	4/30	7/30
Fines fraction	0.29000	0.31651	0.32426	21/30	20/30

MAE is averaged equally over all thirty campaigns in each response’s original scale; errors are not ranked across responses. Recovery, purity and fines are fractions, while size is the declared size index. Mean and Nearest denote the public observation-mean and nearest-neighbour predictors. Wins count strictly lower agent MAE; neither comparison has ties. The source-assay shortfall remains included, and the five world instances are reused across arms and independent budget sessions.

Table E2 accompanies the purity coverage comparison in Figure 6d. Wider intervals need not cover a biased prediction’s reference, while coverage exceeding the nominal level does not by itself establish calibration or sharpness.

Table E2. Purity interval coverage and width by information arm and budget.

Budget	Information arm	Campaigns	Coverage (%)	Mean width (pp)
12	Opaque	5	98.3	6.81
12	Aligned	5	60.0	12.94
12	MisIndexed	5	23.3	15.53
24	Opaque	5	95.0	8.59
24	Aligned	5	73.3	12.37
24	MisIndexed	5	55.0	12.31

The original intervals have nominal 80% coverage and concern the pre-final-assay response target. Coverage and interval width are averaged over each campaign’s twelve queries and then over its five-world arm-budget group. Each row contains sixty query targets but only five world clusters; queries are not independent world replicates. Width is expressed in percentage points of purity. All original intervals and source failures remain unchanged.

F Complete most-dilute equilibrium predictions

F.1 Original intervals and campaign-specific observations

The main-text equilibrium comparison shows point predictions for the most dilute recipe. Table F1 retains all fifteen original 80% prediction intervals and each campaign’s source-response range. These are prediction intervals issued by the agent, not confidence intervals on an estimated mean. Each source range covers twelve final assays; the reference is the mean of the original five evaluator observations and was unavailable during prediction or retrospective reflection.

Table F1. Dissociation predictions at the most dilute recipe, with all values in percent.

World / arm	Source range	Point prediction	80% interval	Reference mean
1 / Opaque	6.18-8.34	69.90	[25.00, 90.00]	71.60
1 / Aligned	6.20-9.23	10.40	[4.40, 22.50]	71.60
1 / MisIndexed	6.11-9.25	10.92	[7.72, 14.12]	71.60
2 / Opaque	6.17-8.57	62.90	[45.00, 80.00]	65.46
2 / Aligned	6.12-8.06	8.04	[5.30, 10.80]	65.46
2 / MisIndexed	6.23-7.98	8.06	[5.56, 10.56]	65.46
3 / Opaque	5.94-8.22	56.30	[43.00, 70.00]	57.97
3 / Aligned	6.34-8.02	11.00	[4.50, 19.00]	57.97
3 / MisIndexed	6.16-7.94	46.00	[10.00, 75.00]	57.97
4 / Opaque	6.19-7.91	52.00	[15.00, 88.00]	51.73
4 / Aligned	5.82-8.14	7.24	[4.00, 10.50]	51.73
4 / MisIndexed	5.81-8.22	9.62	[4.00, 45.00]	51.73
5 / Opaque	5.49-7.48	41.30	[20.00, 70.00]	44.80
5 / Aligned	6.44-7.99	7.90	[3.50, 13.00]	44.80
5 / MisIndexed	6.58-7.79	6.81	[4.00, 9.60]	44.80

The recipe contains 1 micromole in 75 millilitres (nominal concentration 13.3 micromolar). The pooled research-observation band in the main figure is the minimum and maximum across all 180 assays; the ranges above are specific to each campaign. World 3 MisIndexed retains its mixed-explanation departure from the observed plateau. Public explanation categories and K1/Q chronology remain in section C.4.